

BFS Traversal

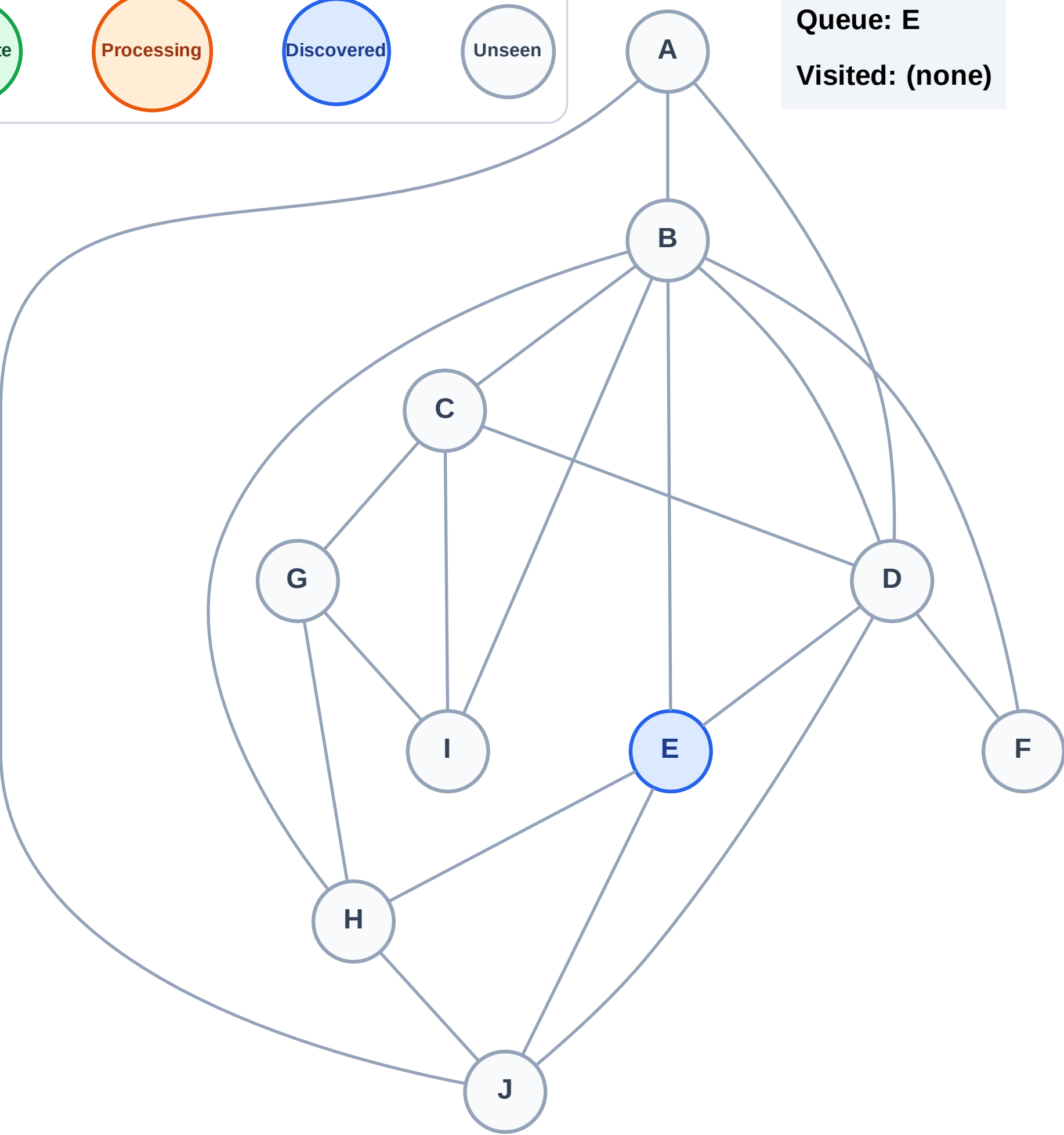
Step 1: Start at E

Legend



Queue: E

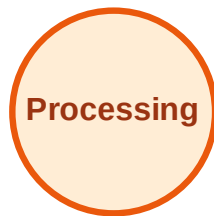
Visited: (none)



BFS Traversal

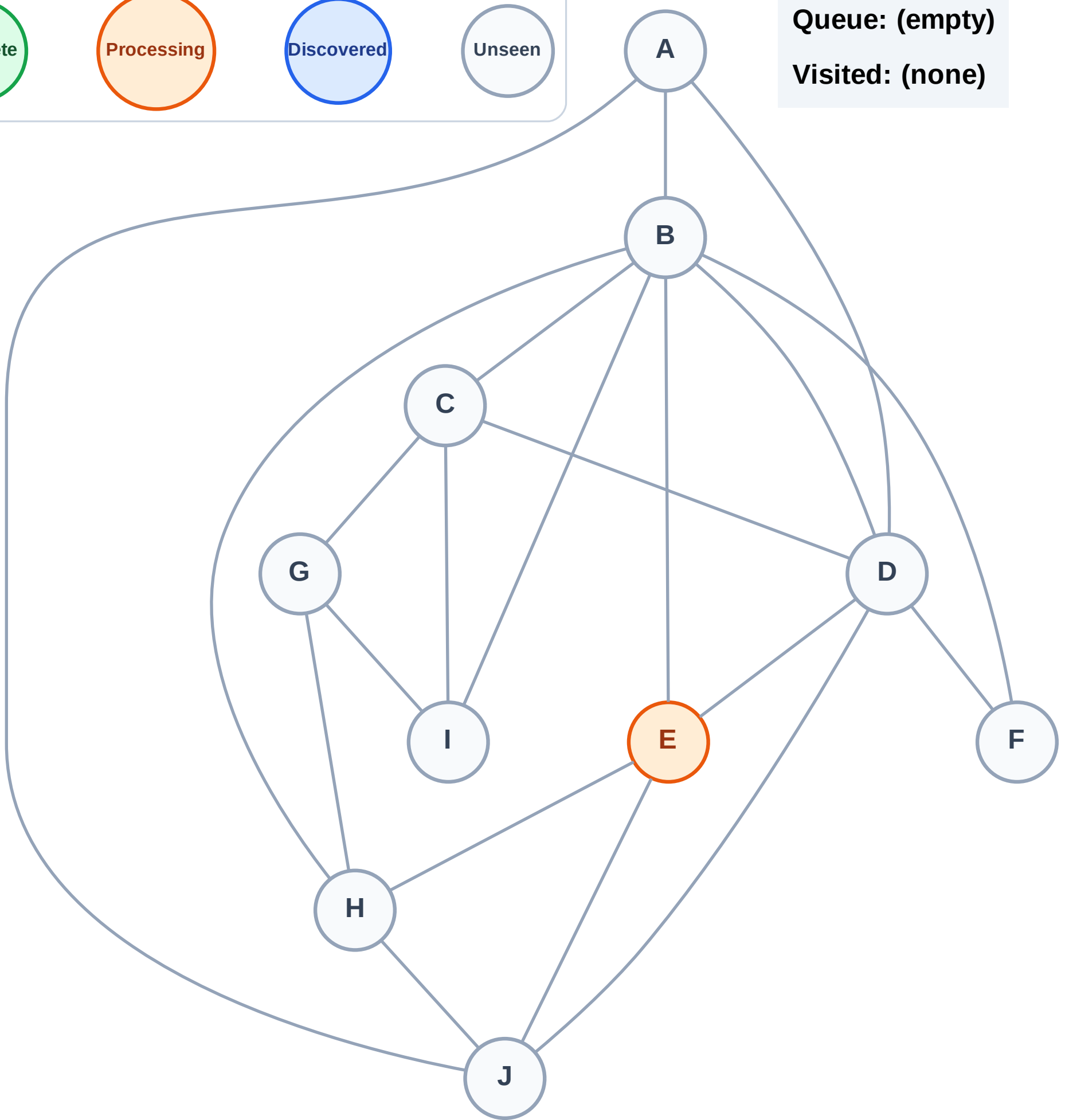
Step 2: Process node E

Legend



Queue: (empty)

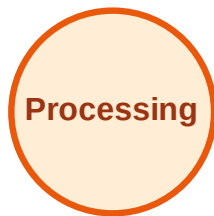
Visited: (none)



BFS Traversal

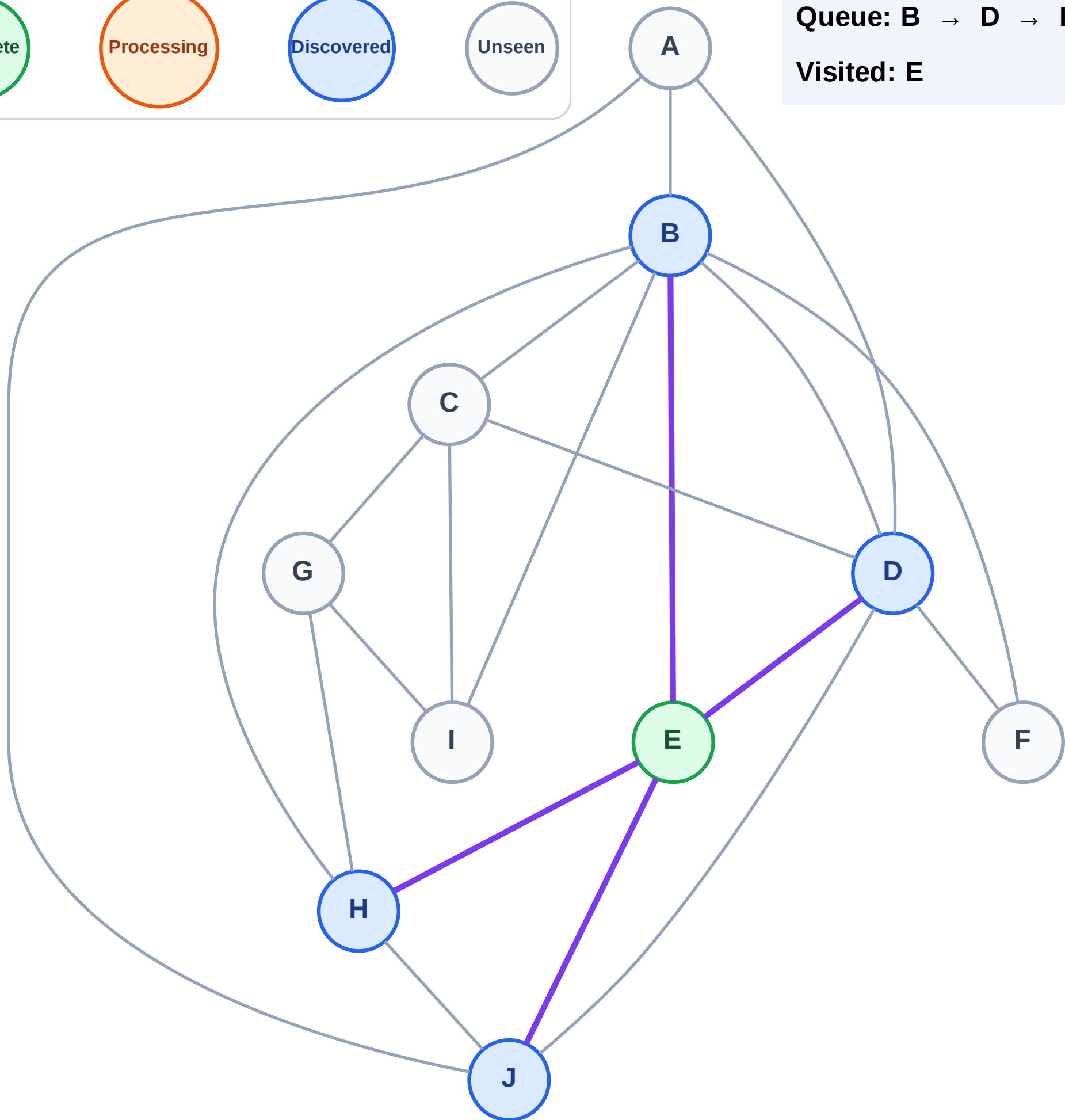
Step 3: Discover B, D, H, J from E

Legend



Queue: B → D → H → J

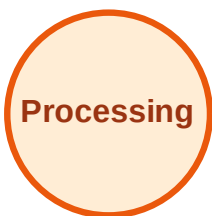
Visited: E



BFS Traversal

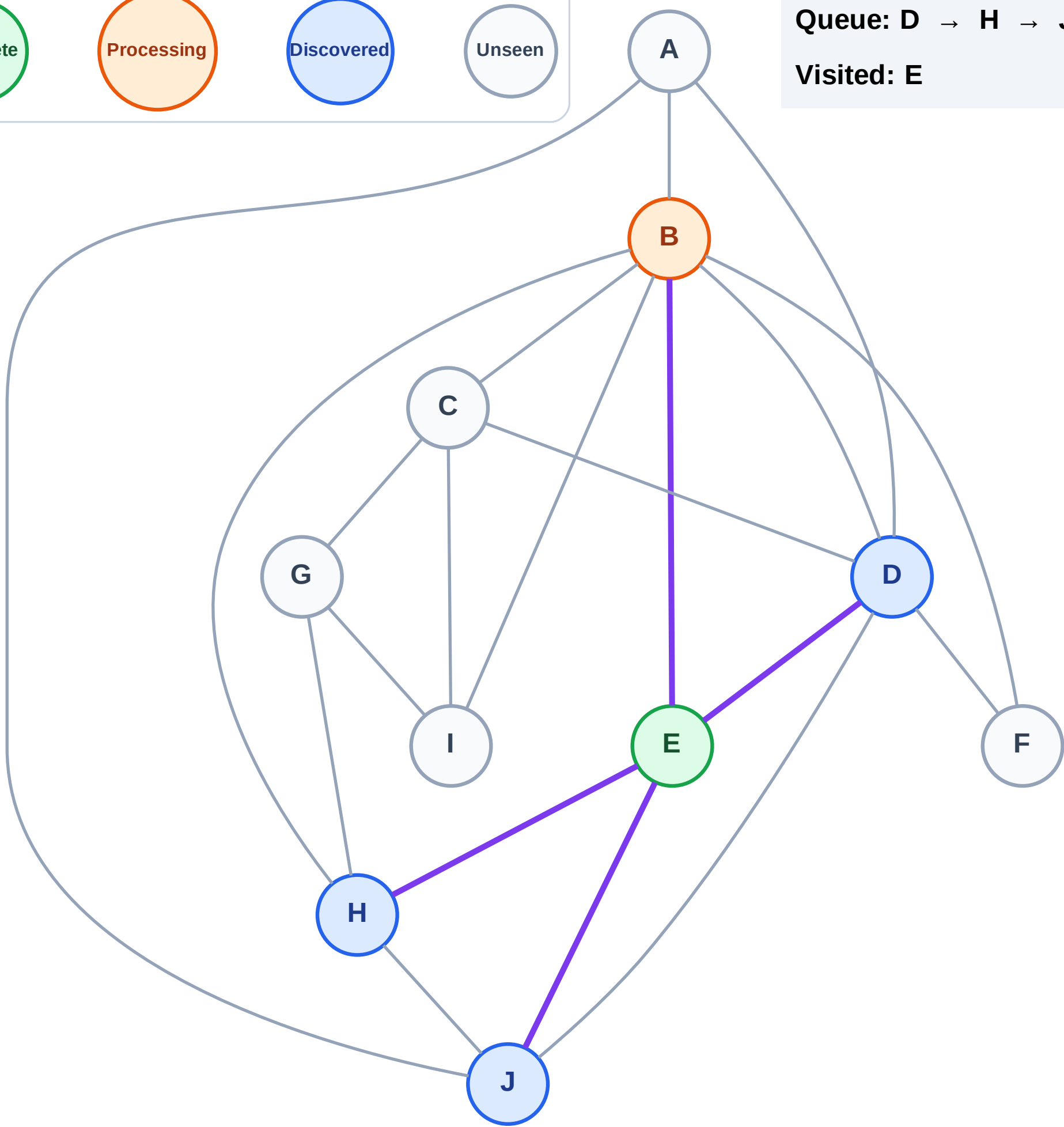
Step 4: Process node B

Legend



Queue: D → H → J

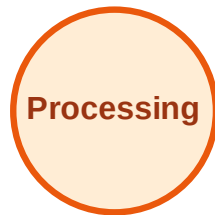
Visited: E



BFS Traversal

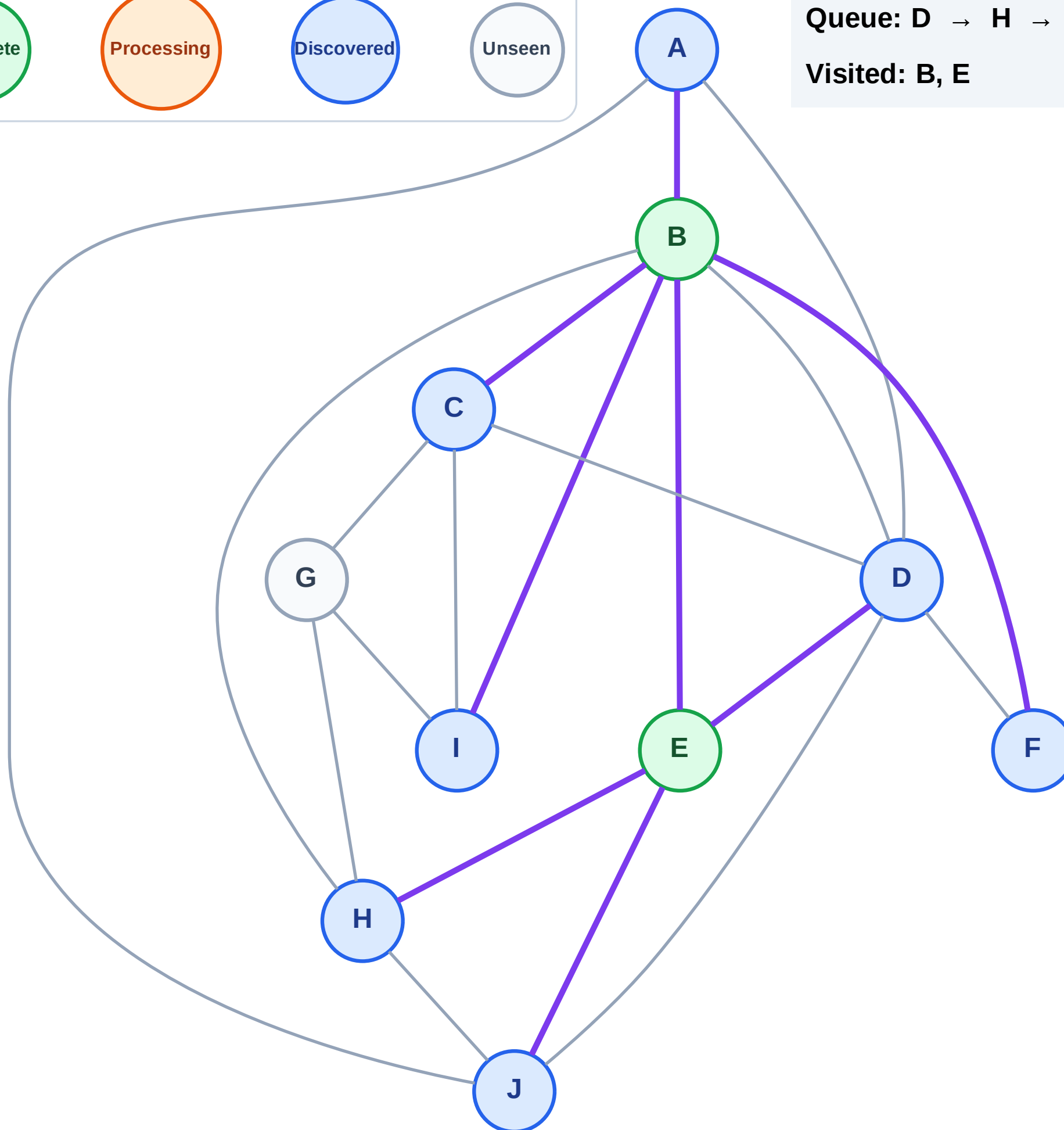
Step 5: Discover A, C, F, I from B

Legend



Queue: D → H → J → A → C → F → I

Visited: B, E



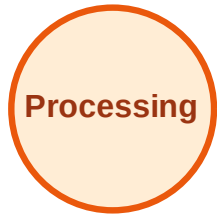
BFS Traversal

Step 6: Process node D

Legend



Complete



Processing

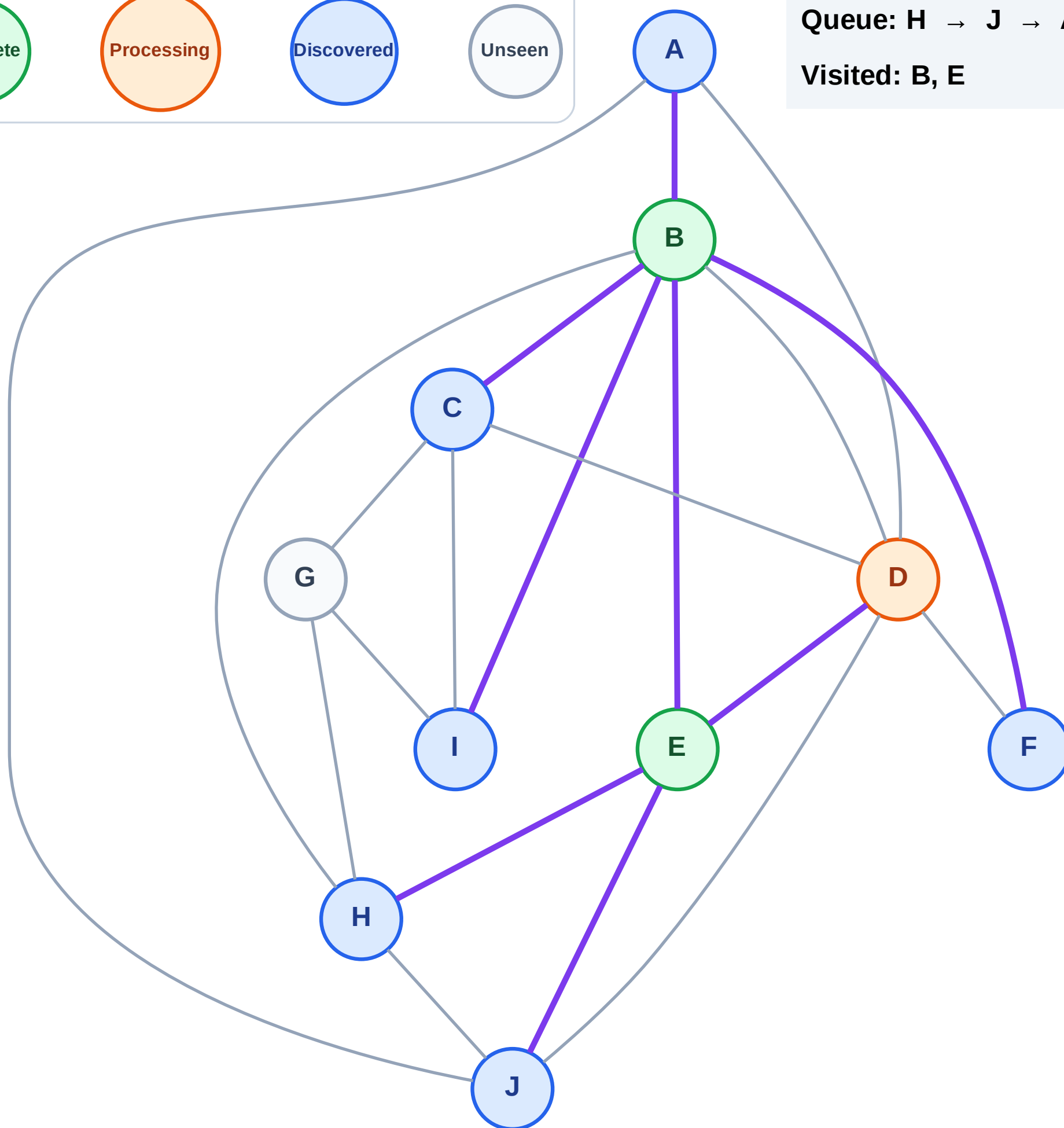


Discovered



Unseen

Queue: H → J → A → C → F → I
Visited: B, E



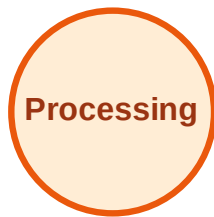
BFS Traversal

Step 7: No new neighbors from D

Legend



Complete



Processing

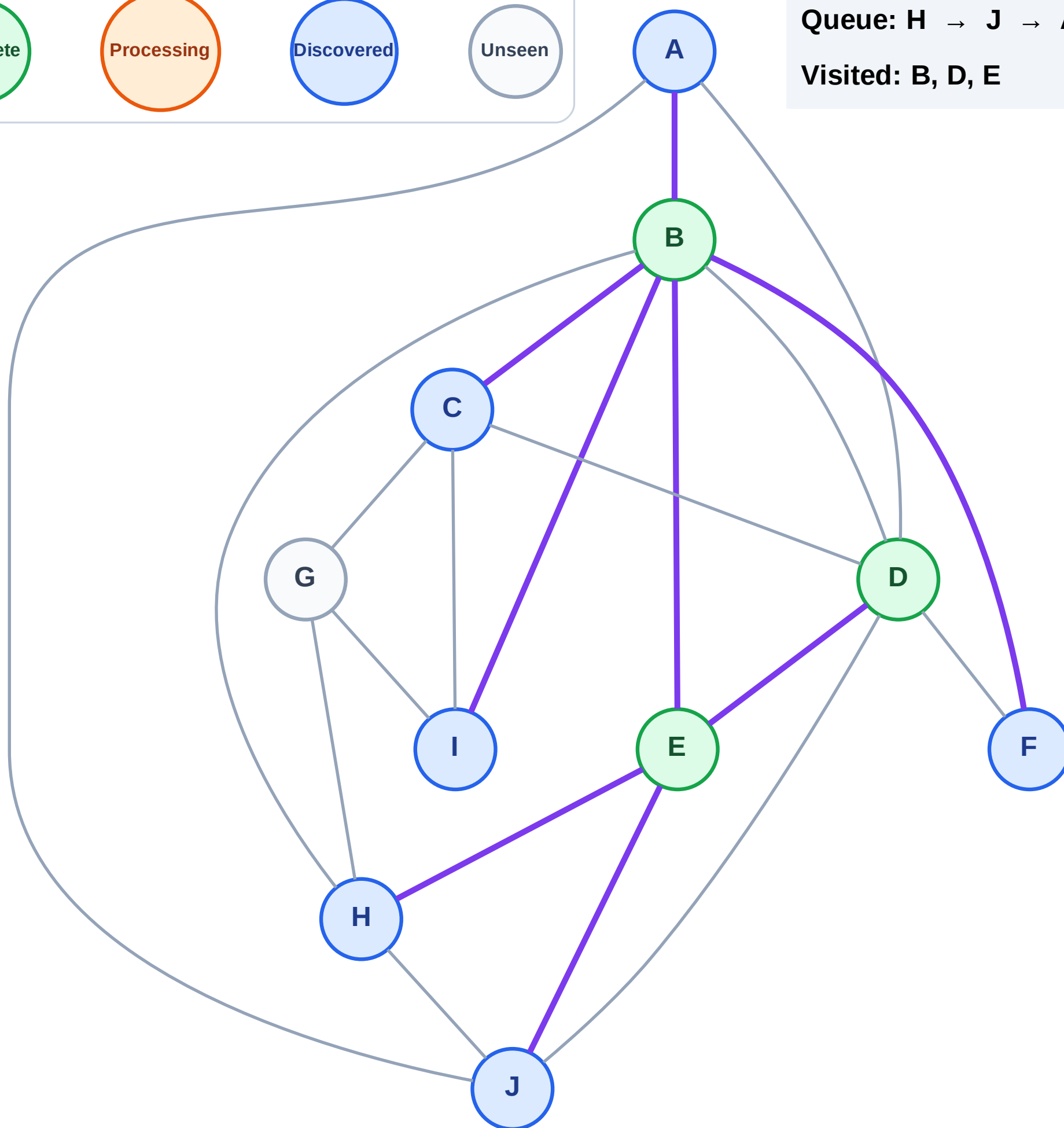


Discovered



Unseen

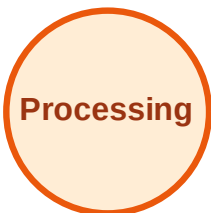
Queue: H → J → A → C → F → I
Visited: B, D, E



BFS Traversal

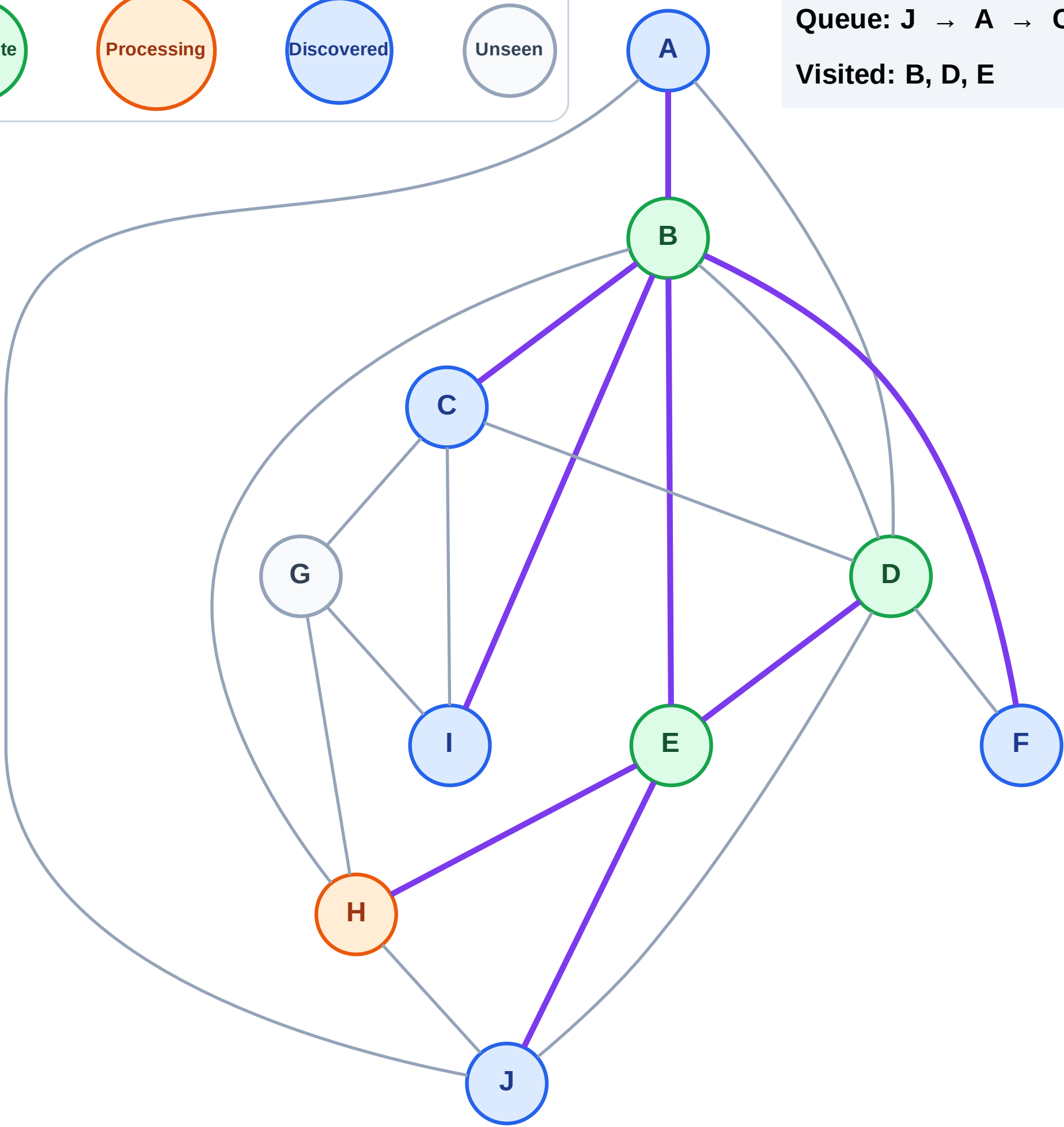
Step 8: Process node H

Legend



Queue: J → A → C → F → I

Visited: B, D, E



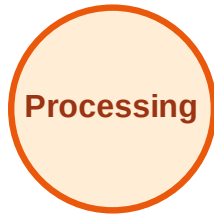
BFS Traversal

Step 9: Discover G from H

Legend



Complete



Processing

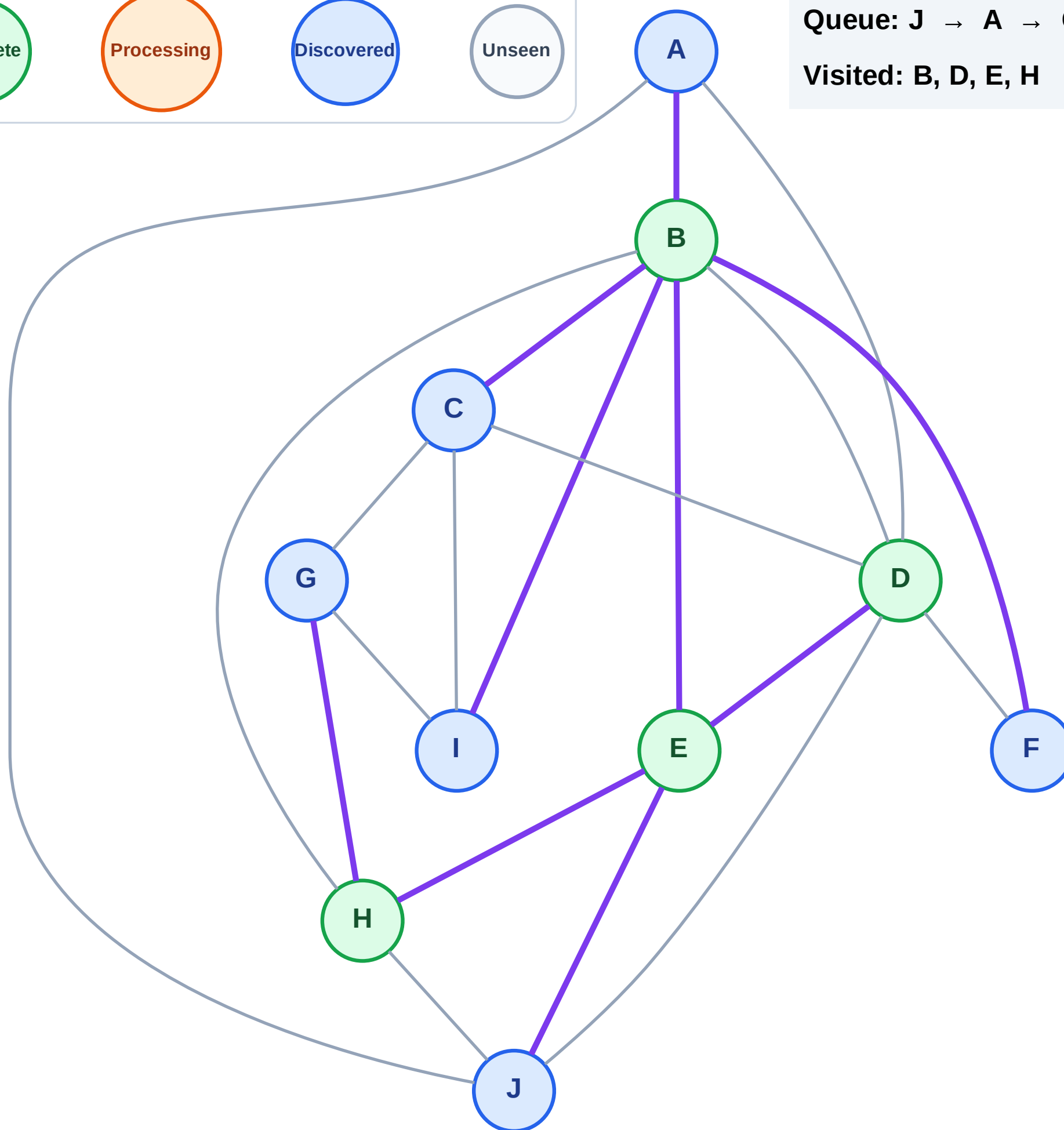


Discovered



Unseen

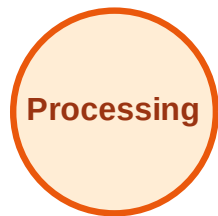
Queue: J → A → C → F → I → G
Visited: B, D, E, H



BFS Traversal

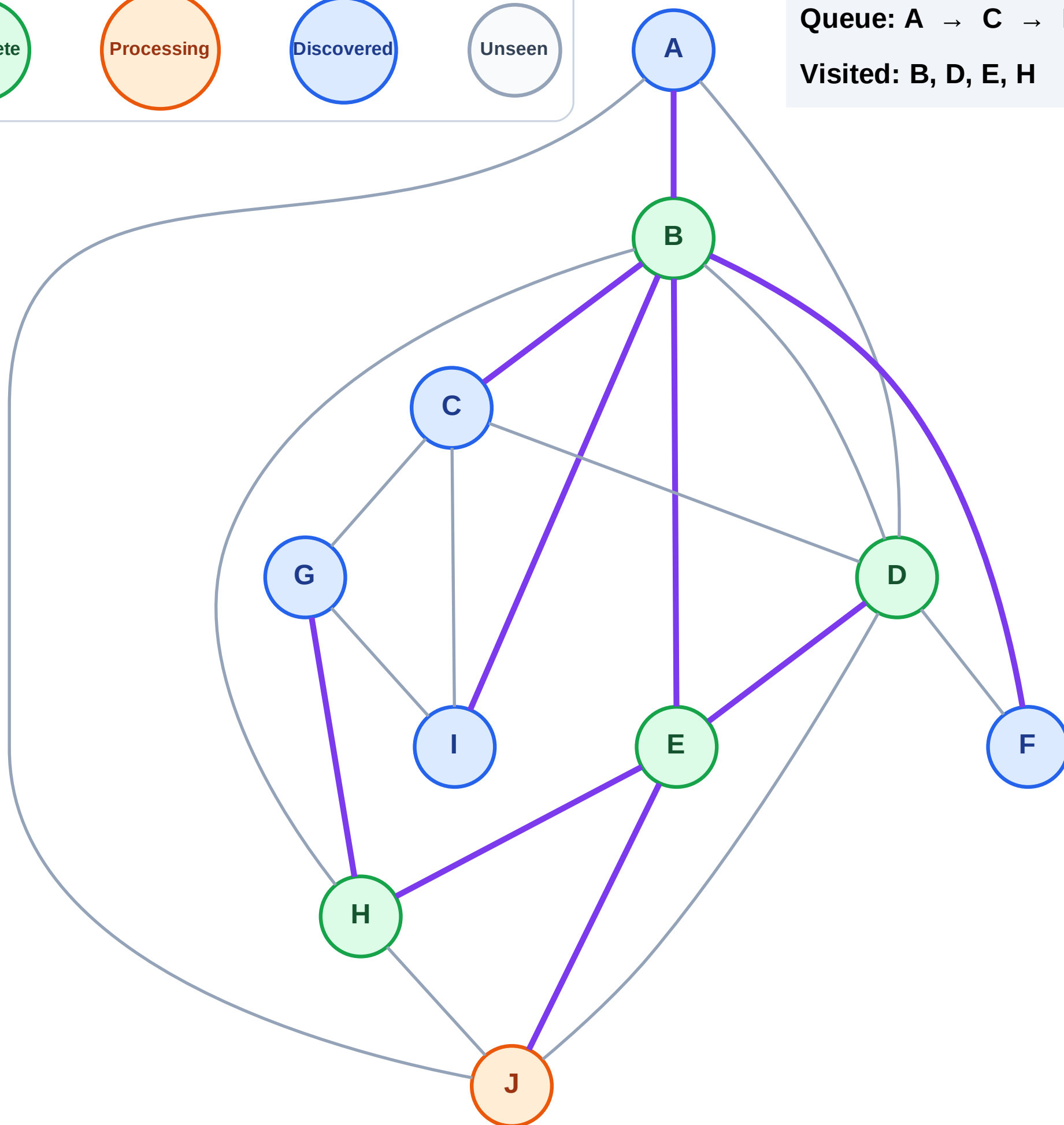
Step 10: Process node J

Legend



Queue: A → C → F → I → G

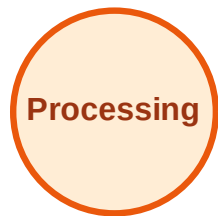
Visited: B, D, E, H



BFS Traversal

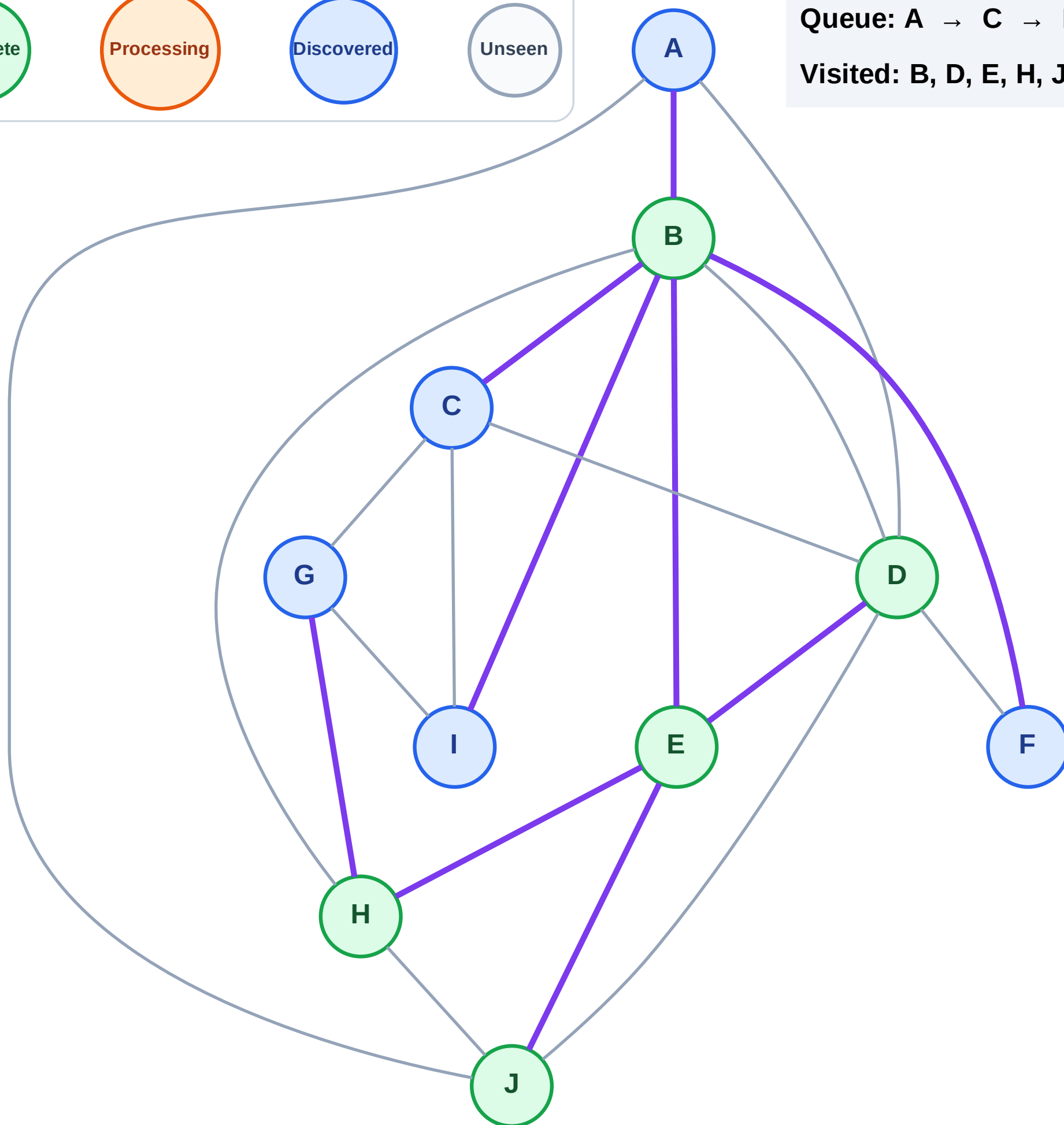
Step 11: No new neighbors from J

Legend



Queue: A → C → F → I → G

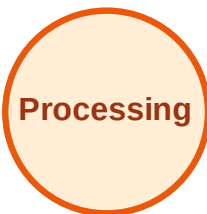
Visited: B, D, E, H, J



BFS Traversal

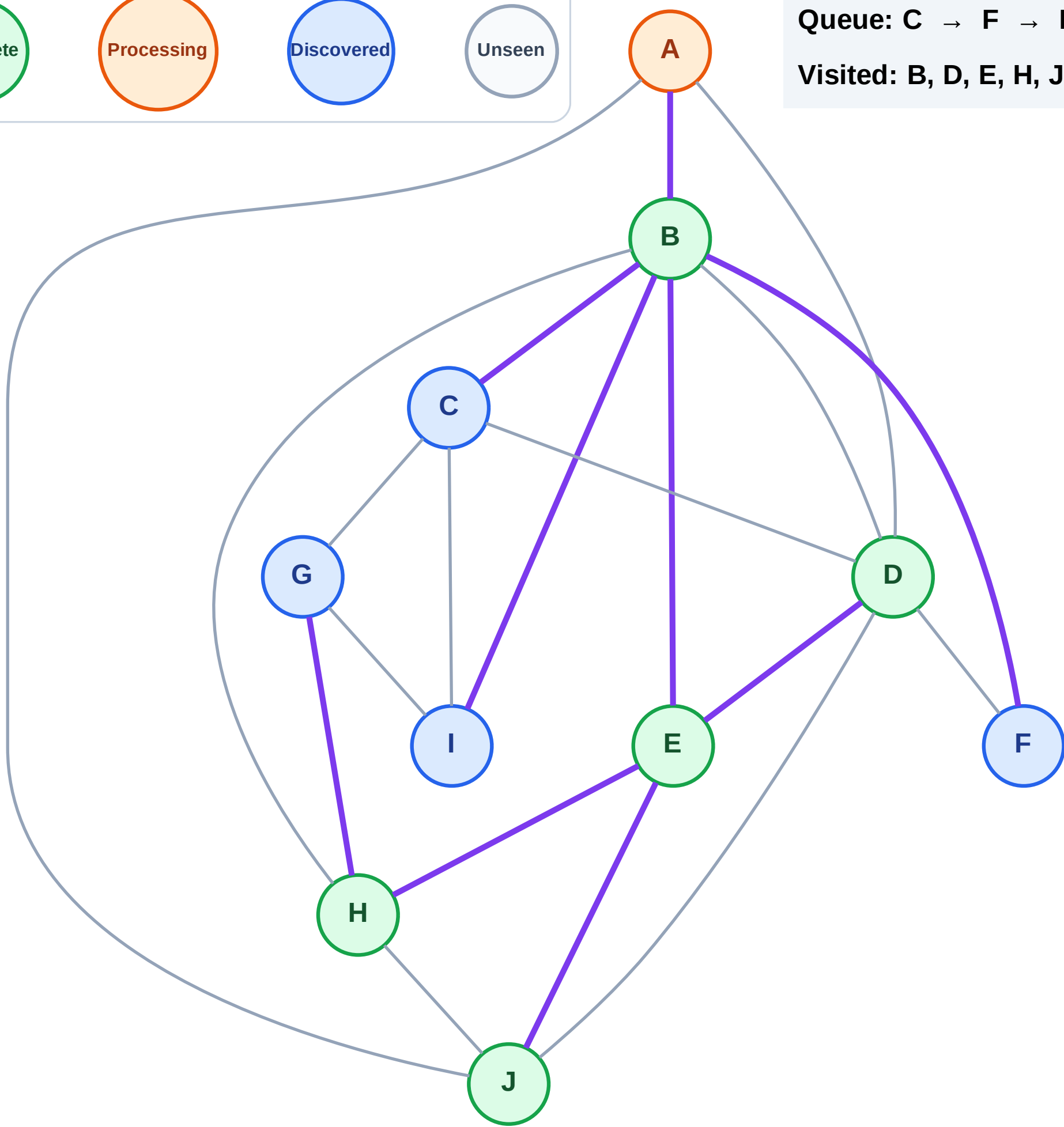
Step 12: Process node A

Legend



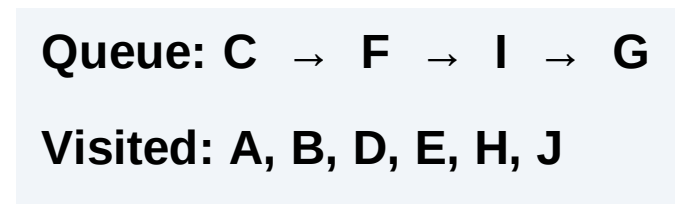
Queue: C → F → I → G

Visited: B, D, E, H, J



Step 13: No new neighbors from A

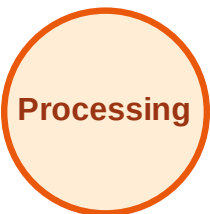
Step 13: No new neighbors from A



BFS Traversal

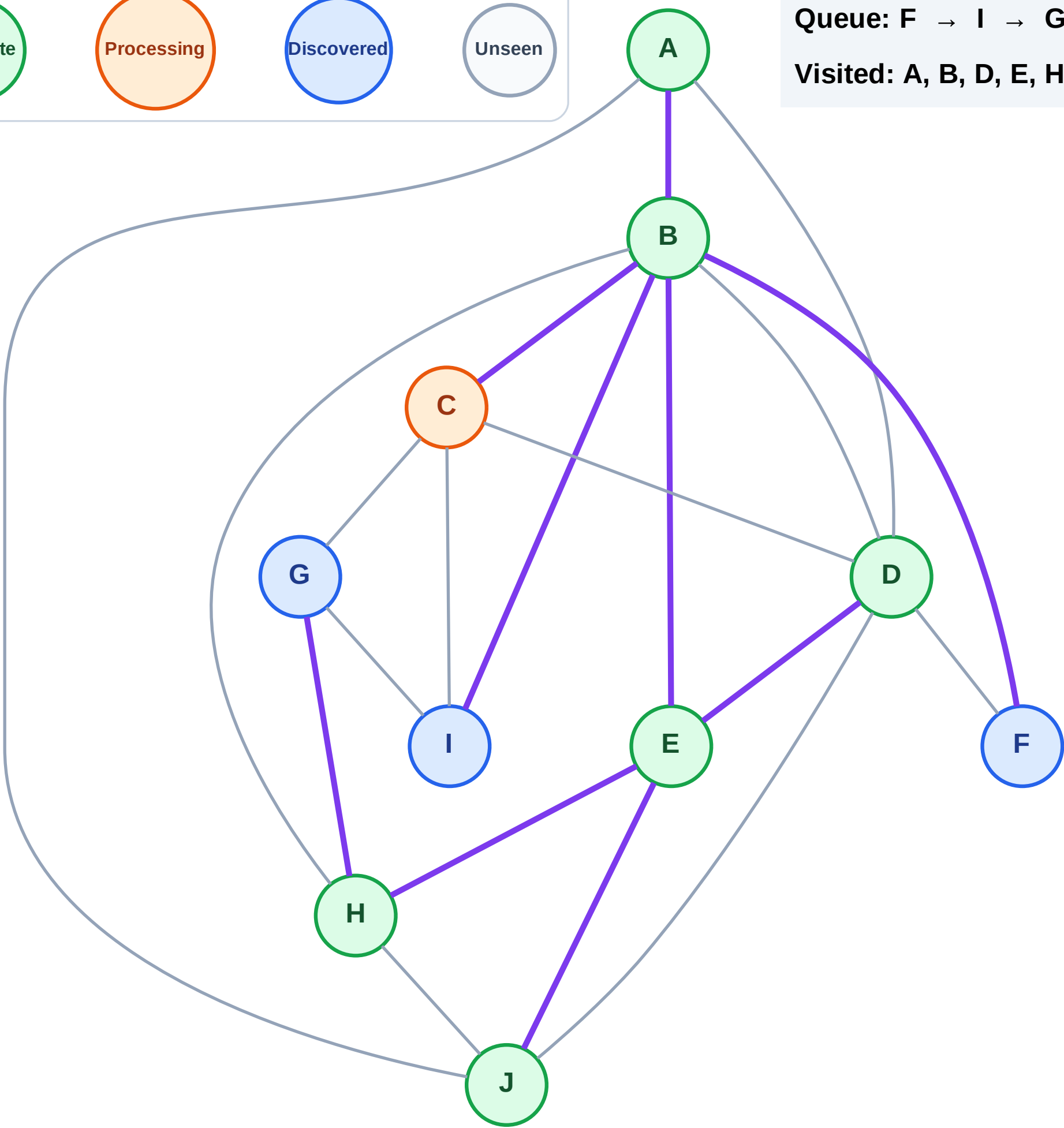
Step 14: Process node C

Legend



Queue: F → I → G

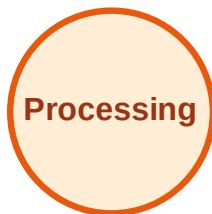
Visited: A, B, D, E, H, J



BFS Traversal

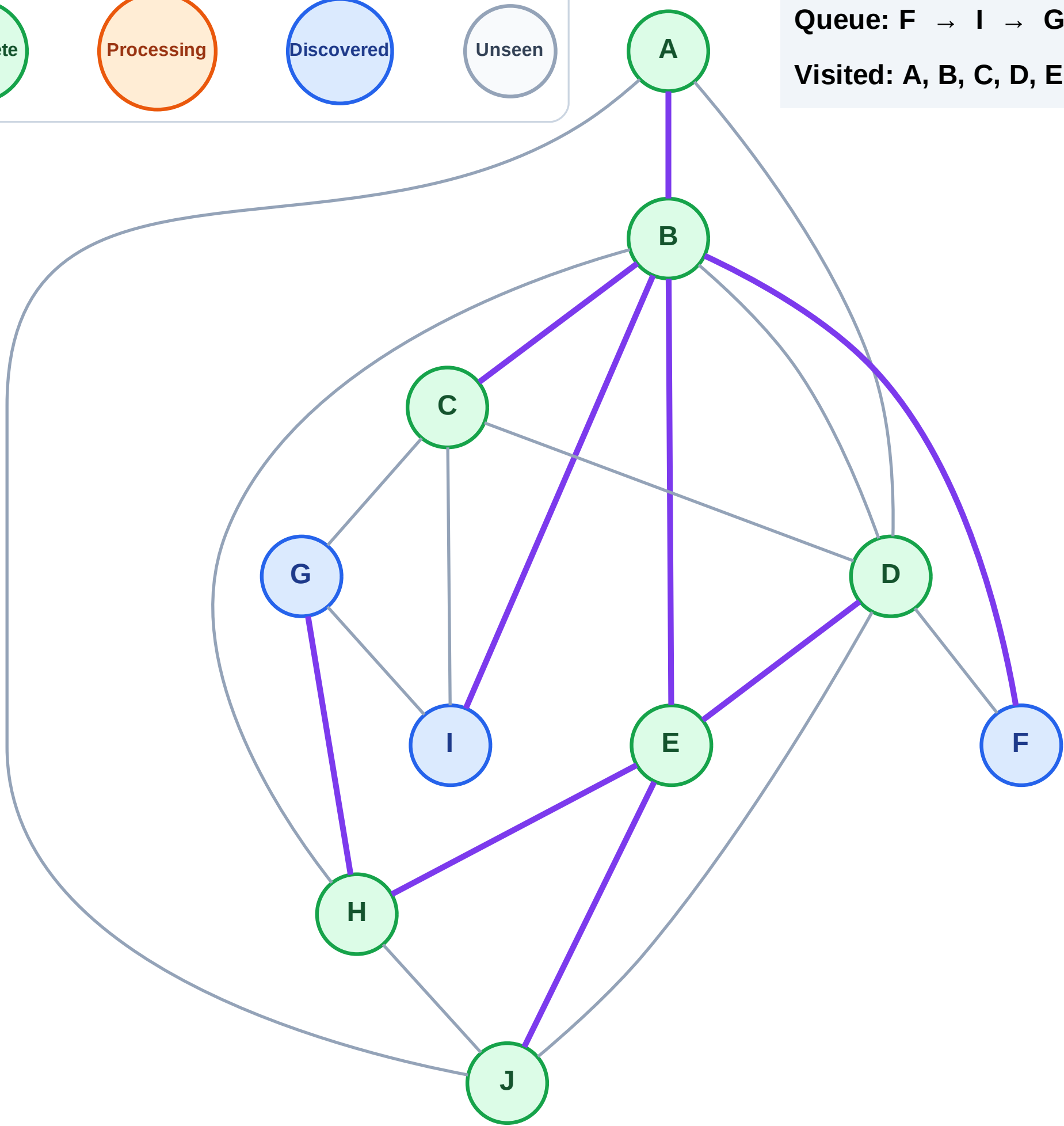
Step 15: No new neighbors from C

Legend



Queue: F → I → G

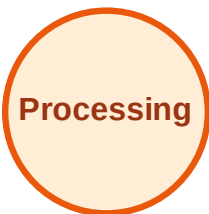
Visited: A, B, C, D, E, H, J



BFS Traversal

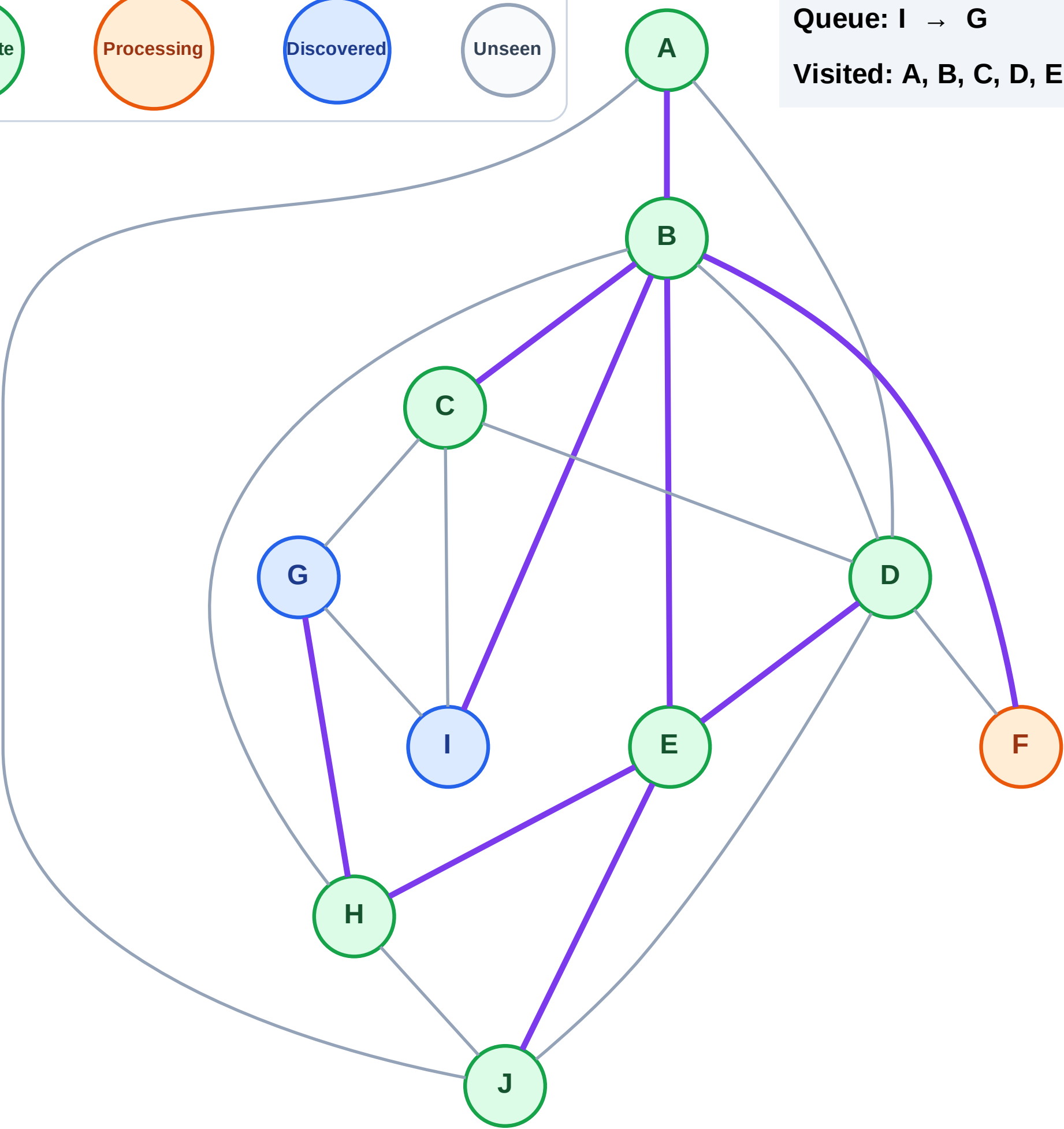
Step 16: Process node F

Legend



Queue: I → G

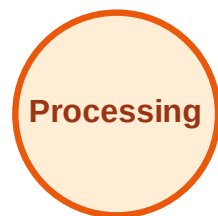
Visited: A, B, C, D, E, H, J



BFS Traversal

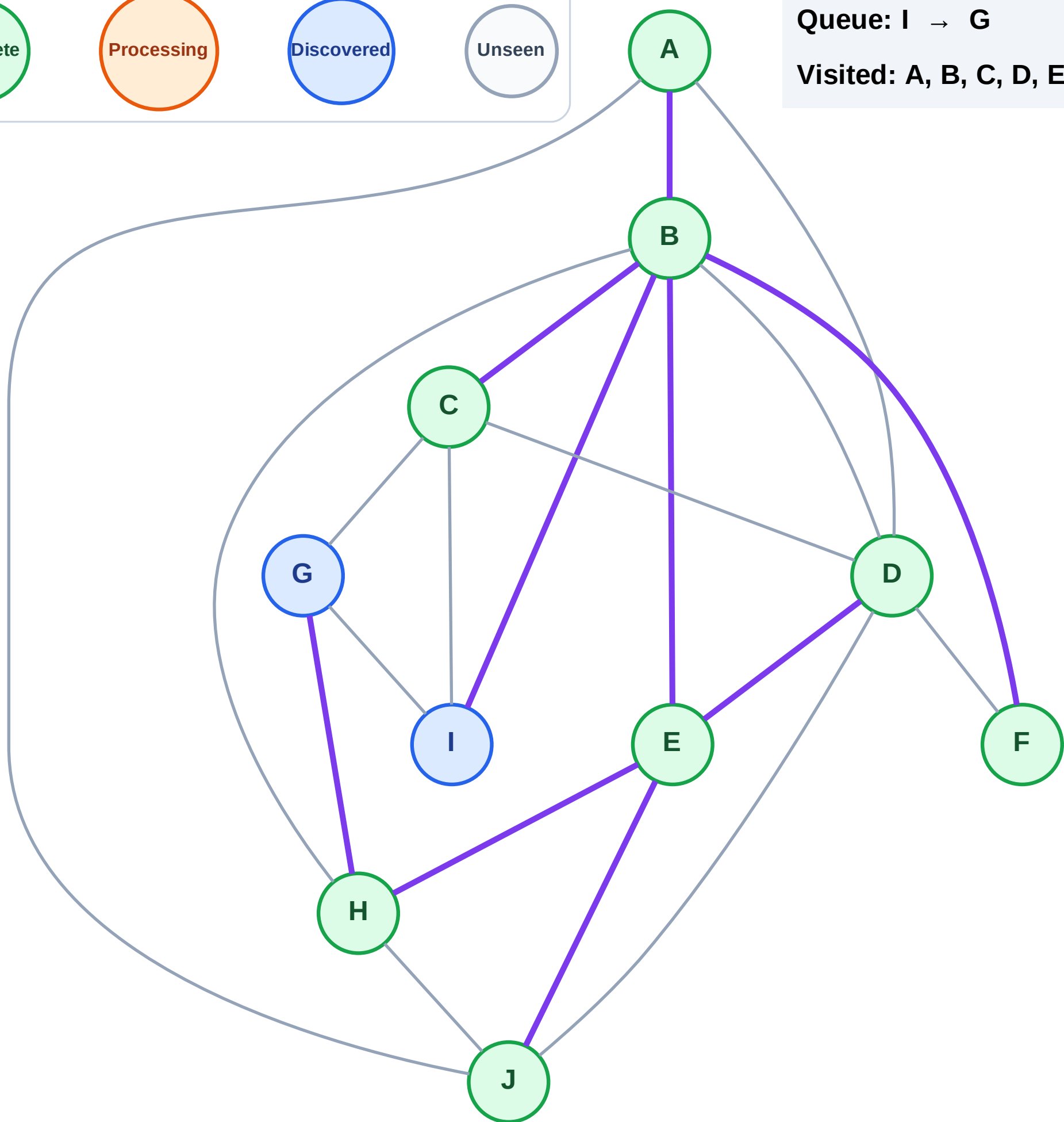
Step 17: No new neighbors from F

Legend



Queue: I → G

Visited: A, B, C, D, E, F, H, J



BFS Traversal

Step 18: Process node I

Legend

Complete

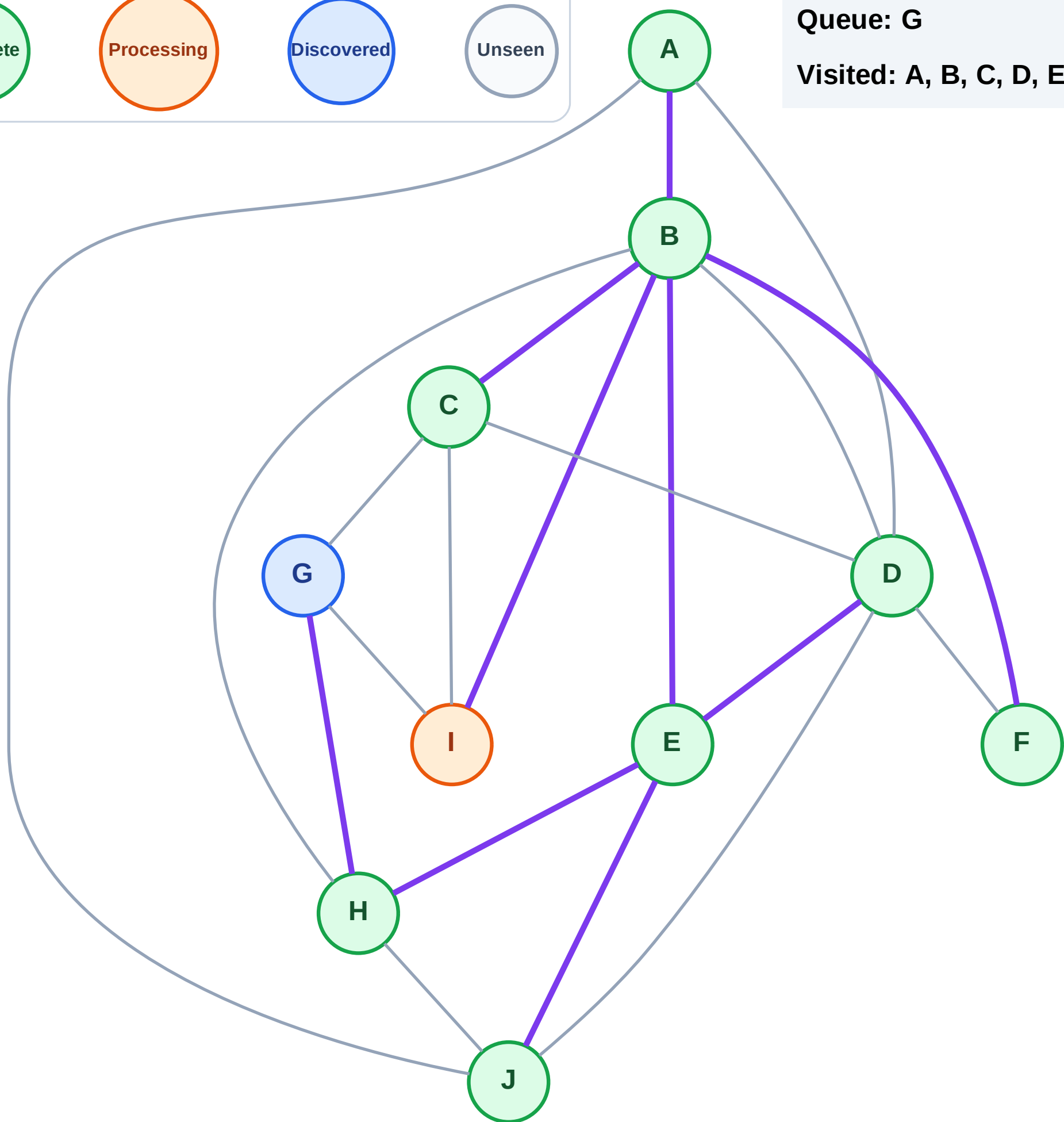
Processing

Discovered

Unseen

Queue: G

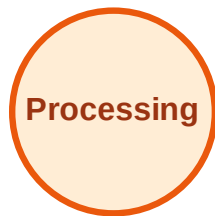
Visited: A, B, C, D, E, F, H, J



BFS Traversal

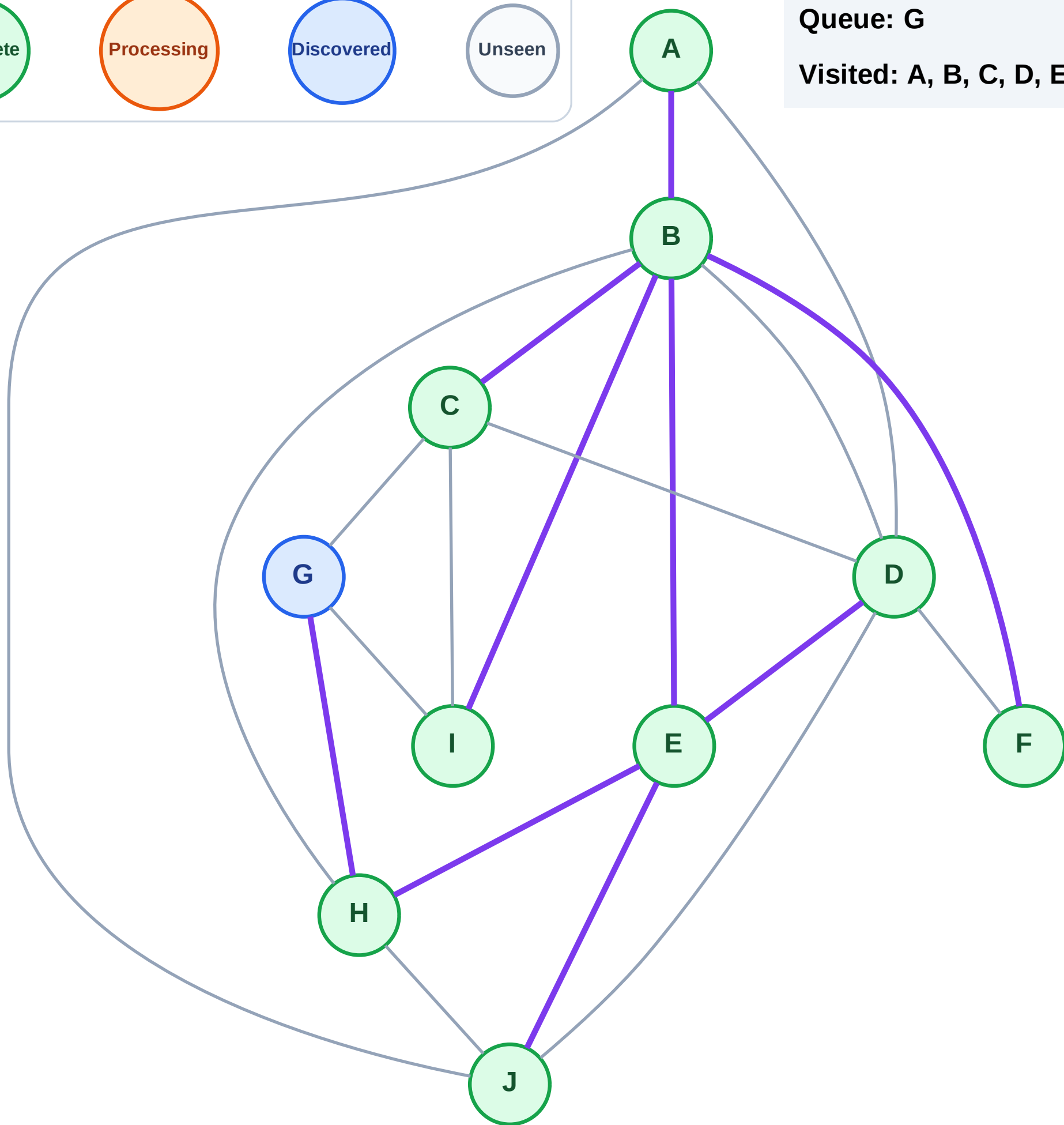
Step 19: No new neighbors from I

Legend



Queue: G

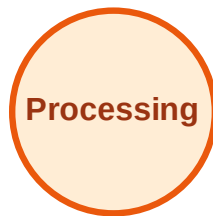
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

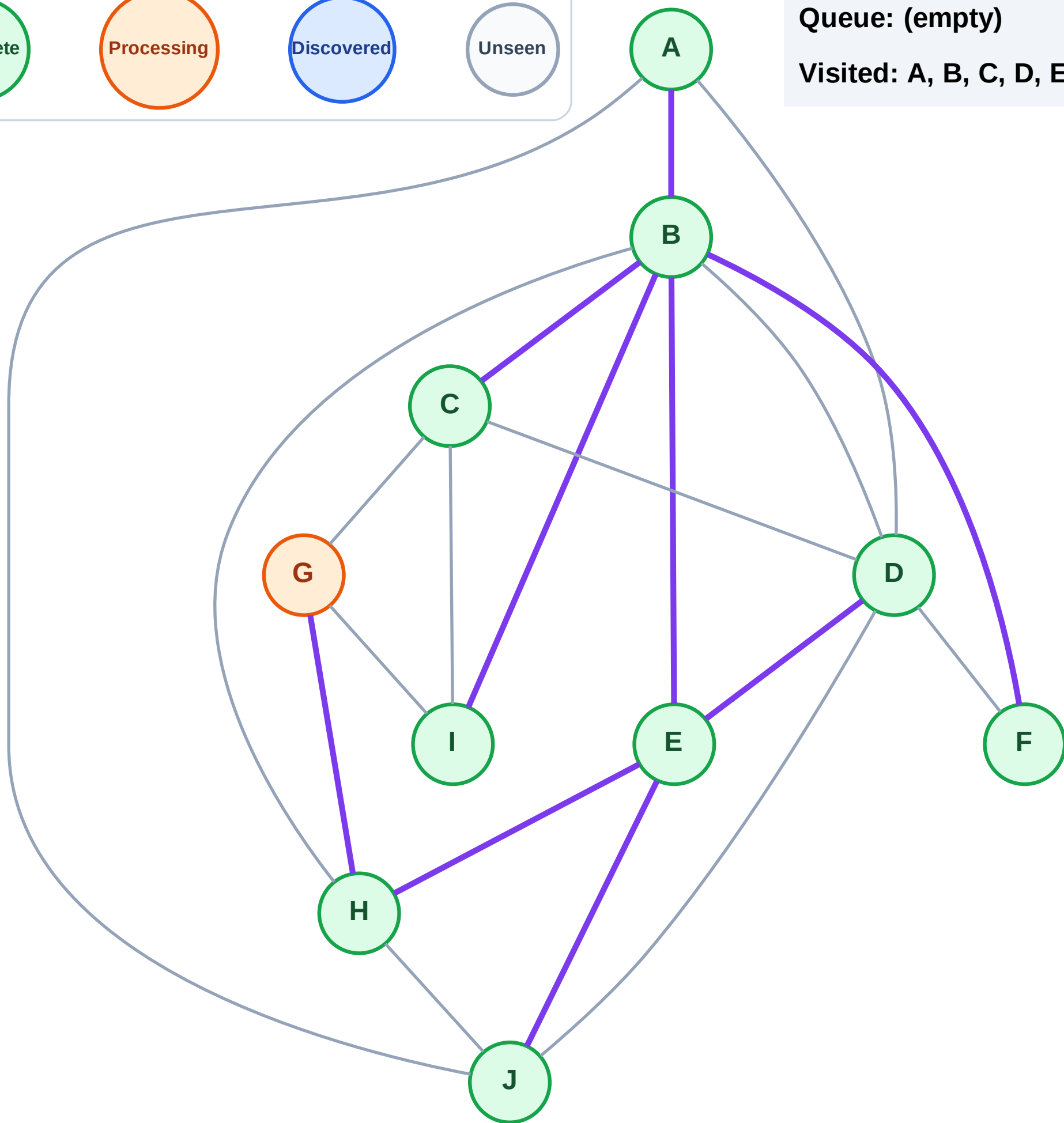
Step 20: Process node G

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

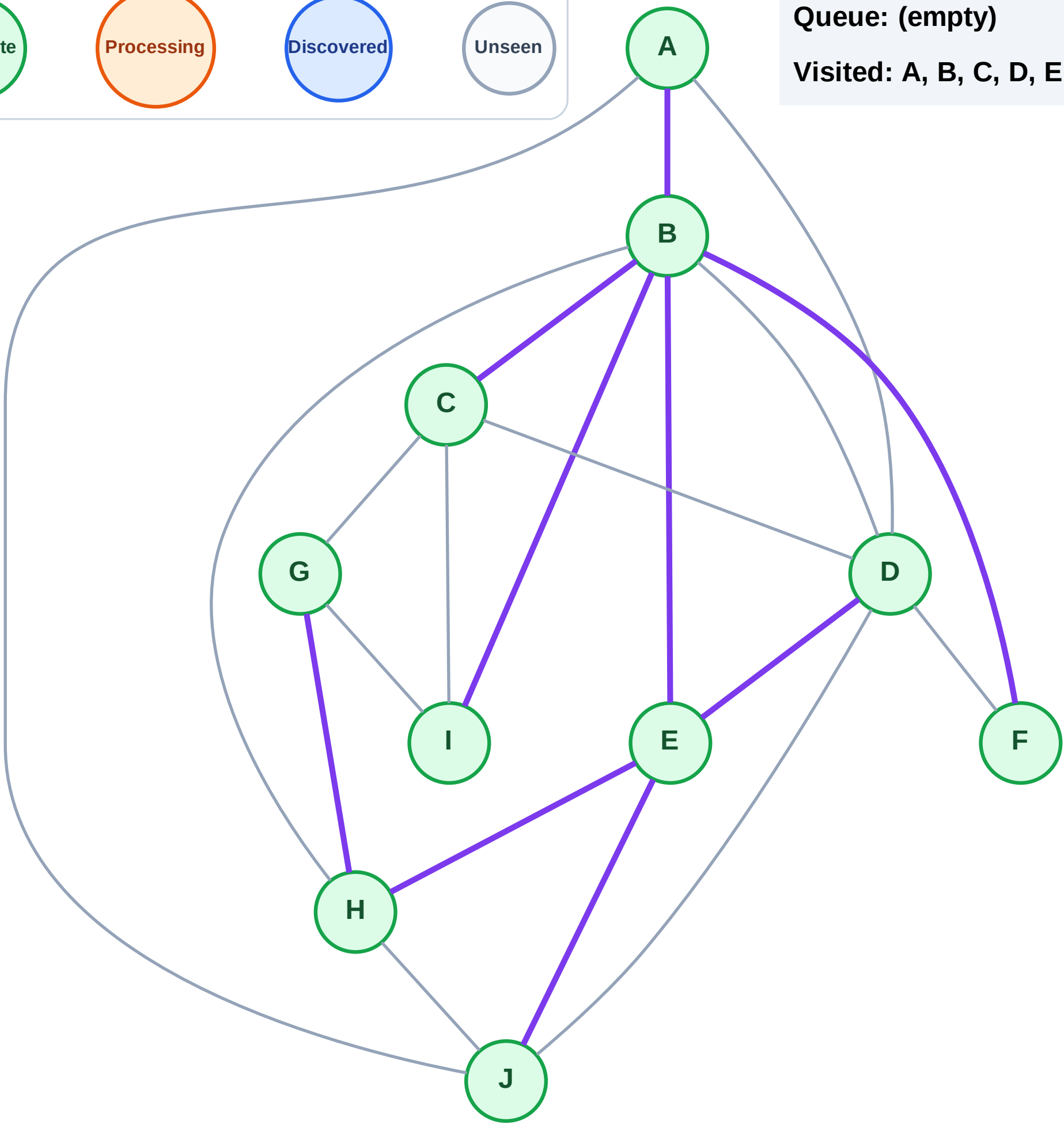
Step 21: No new neighbors from G

Legend



Queue: (empty)

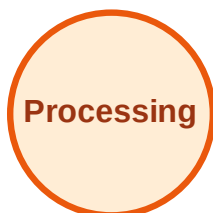
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

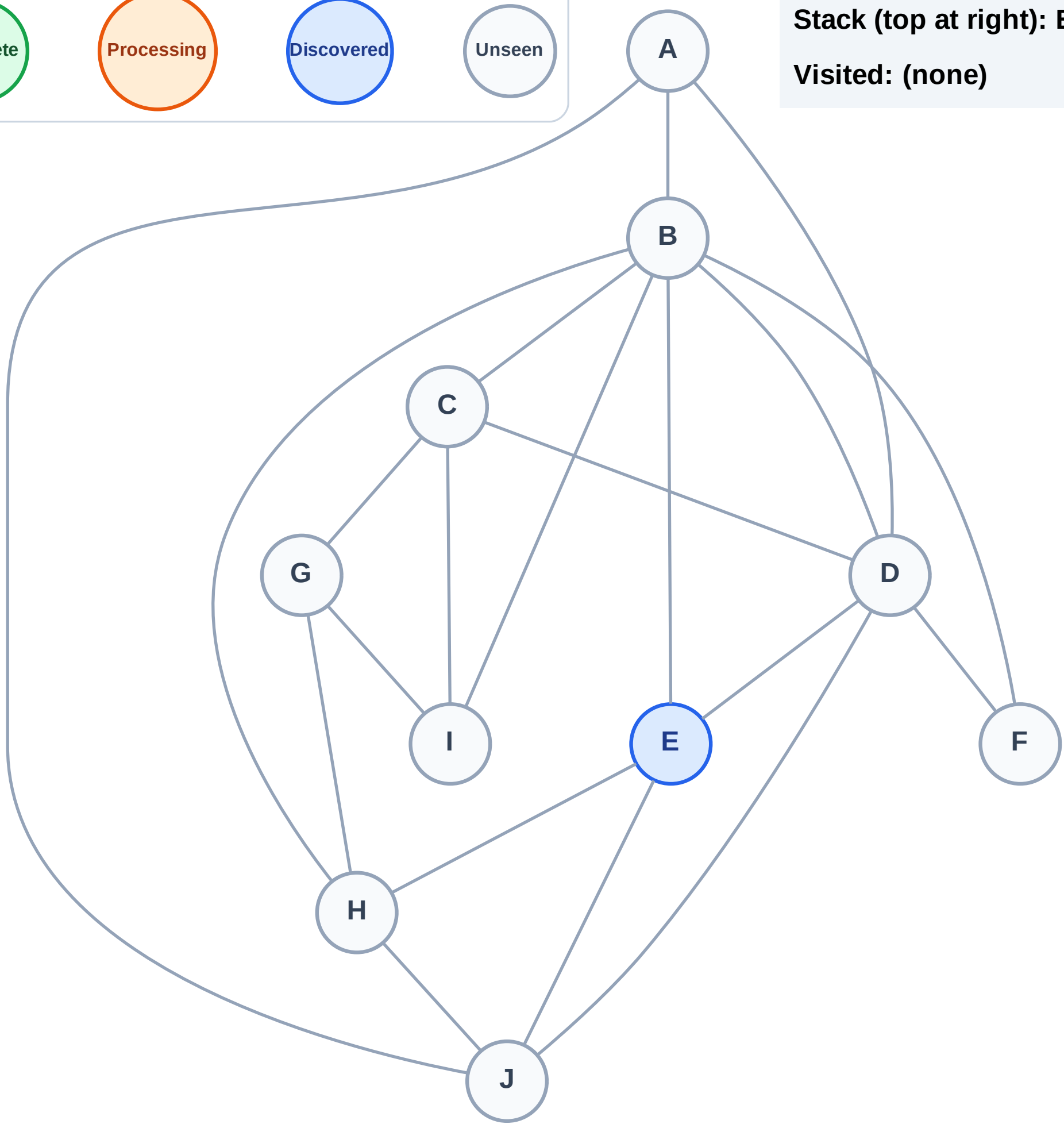
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

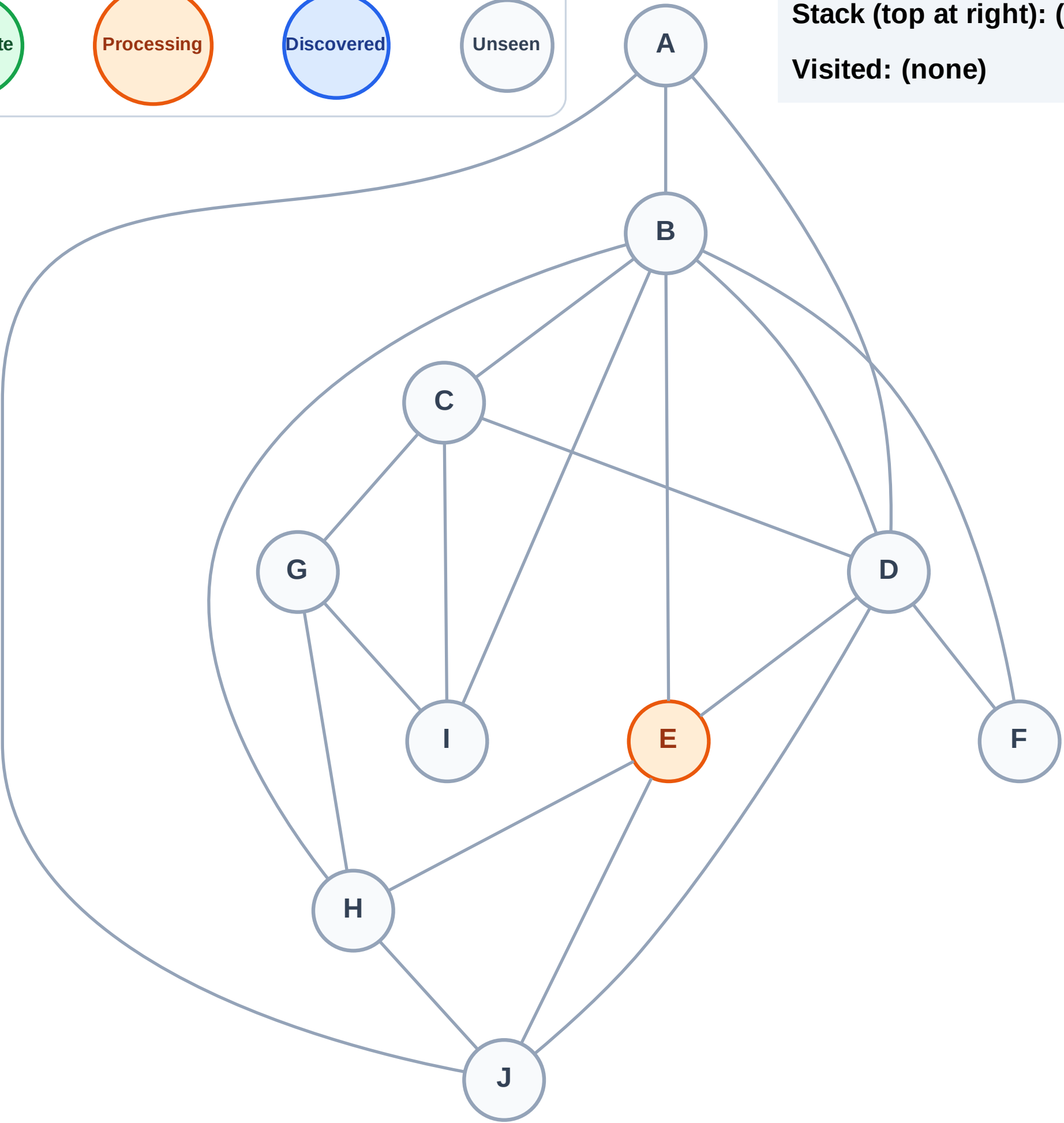
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover J, H, D, B from E

Legend

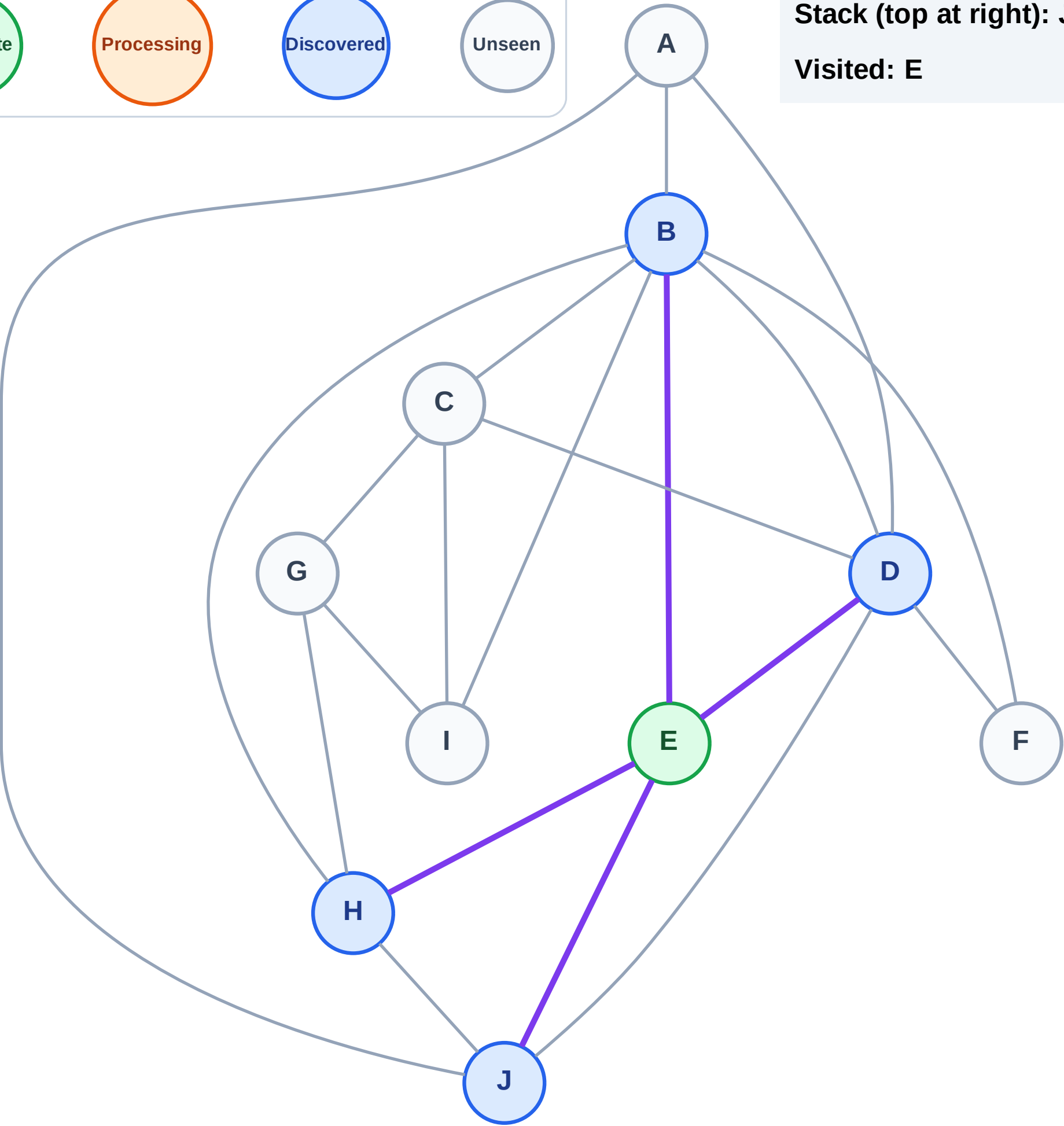
Complete

Processing

Discovered

Unseen

Stack (top at right): J | H | D | B
Visited: E



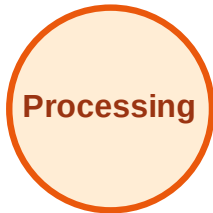
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



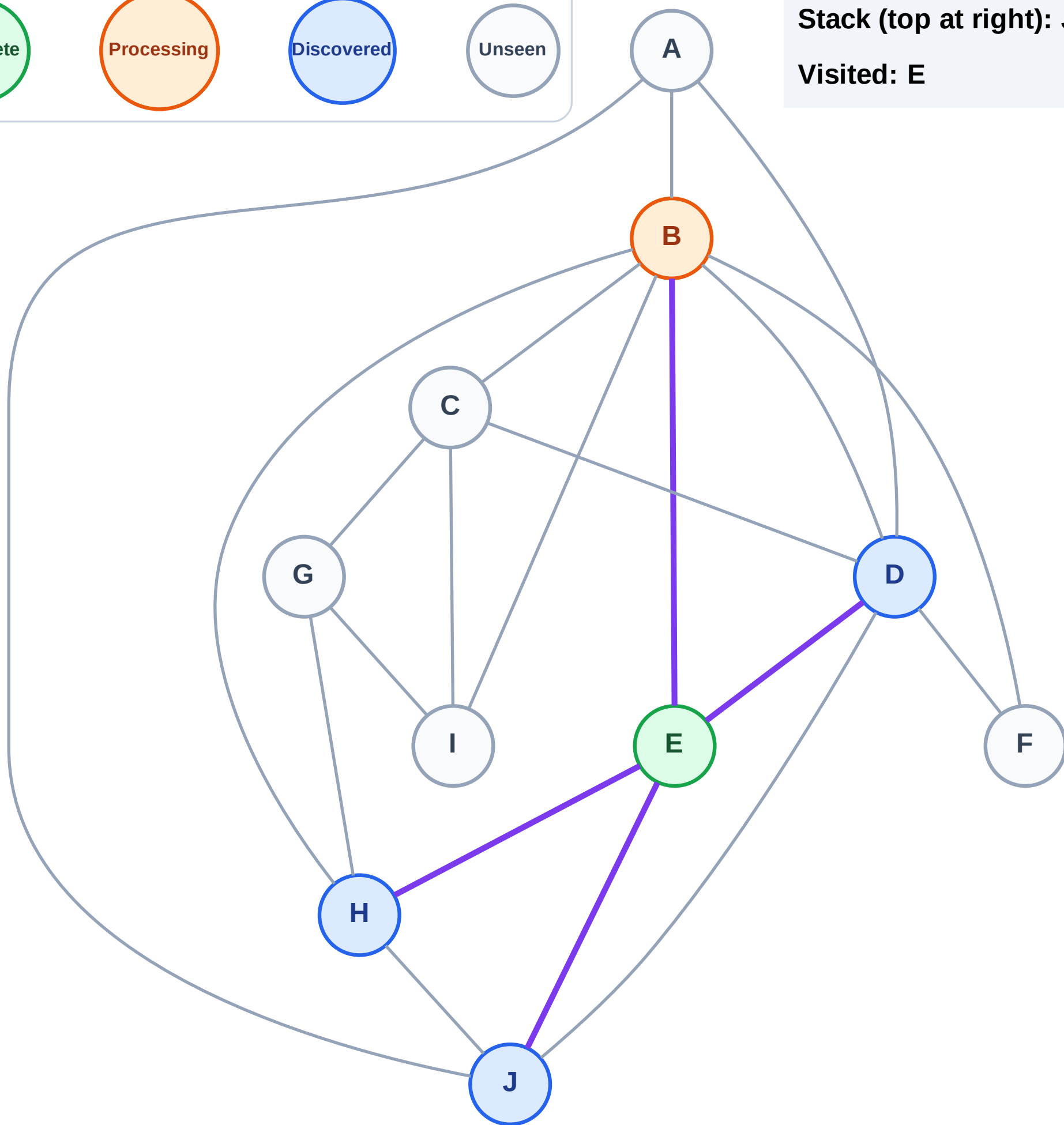
Discovered



Unseen

Stack (top at right): J | H | D

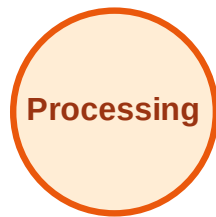
Visited: E



DFS Traversal

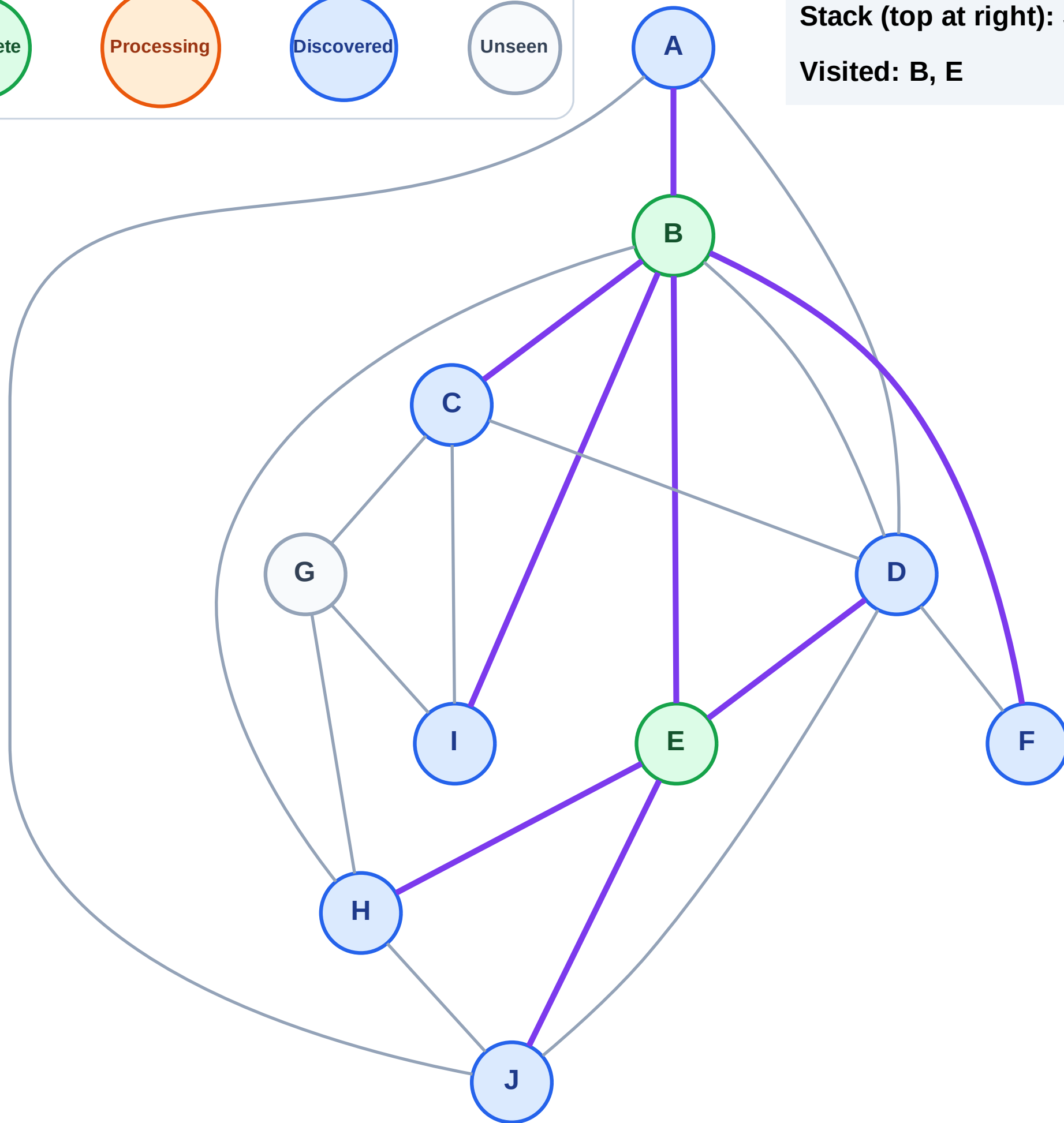
Step 5: Discover I, F, C, A from B

Legend



Stack (top at right): J | H | D | I | F | C | A

Visited: B, E



DFS Traversal

Step 6: Process node A

Legend

Complete

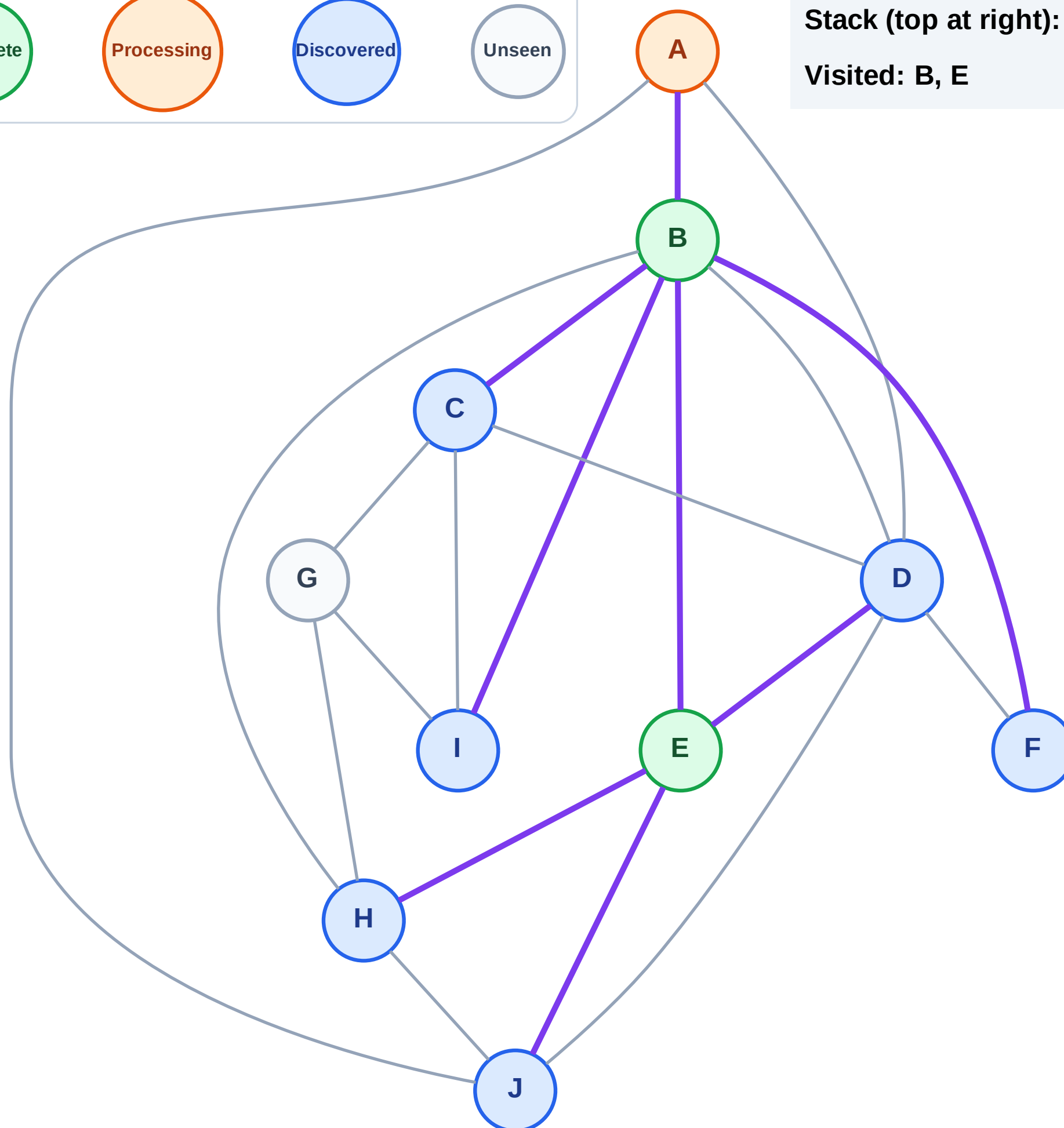
Processing

Discovered

Unseen

Stack (top at right): J | H | D | I | F | C

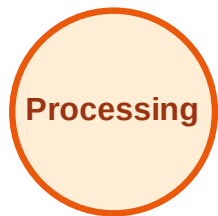
Visited: B, E



DFS Traversal

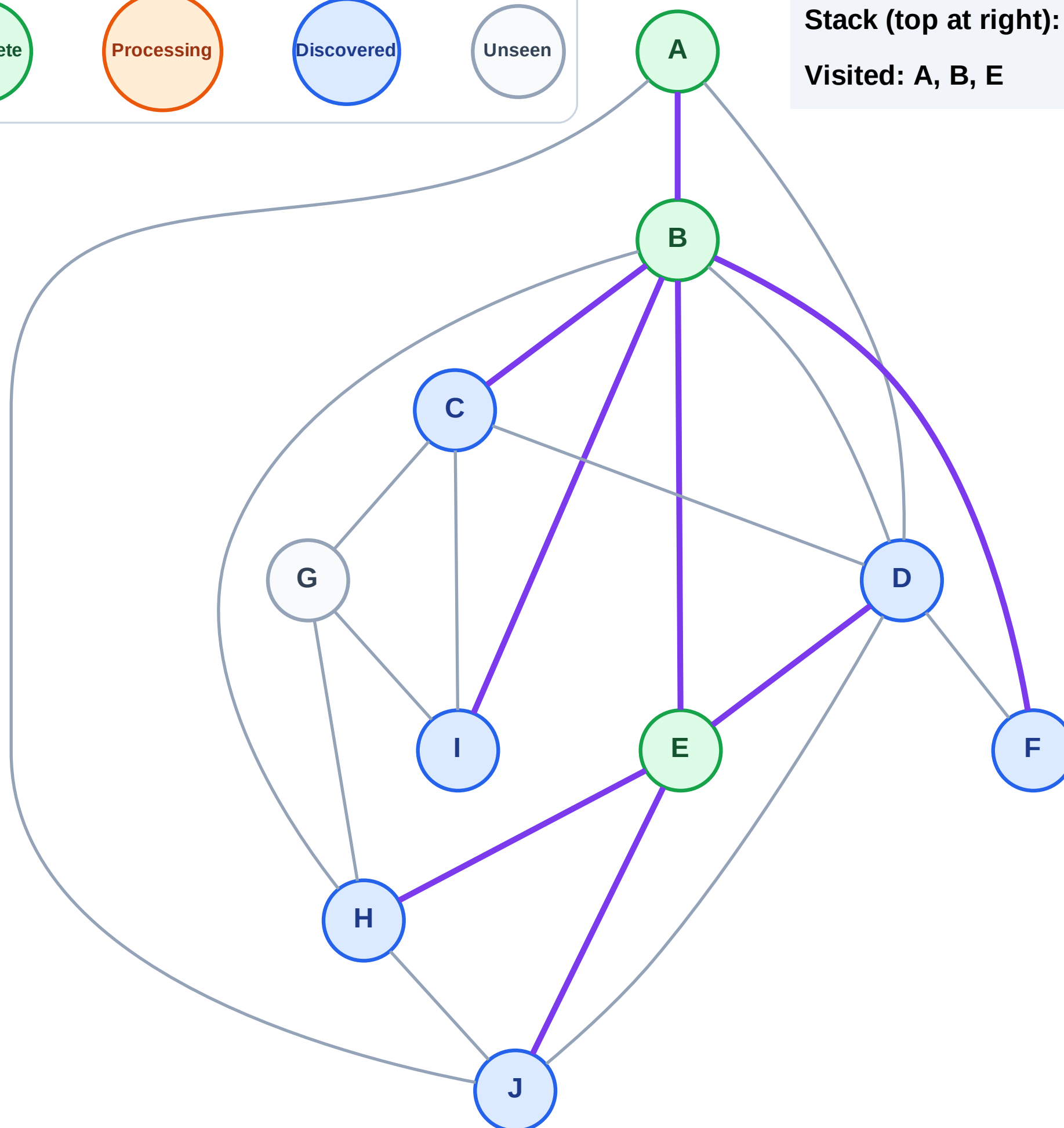
Step 7: No new neighbors from A

Legend



Stack (top at right): J | H | D | I | F | C

Visited: A, B, E



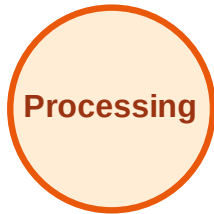
DFS Traversal

Step 8: Process node C

Legend



Complete



Processing



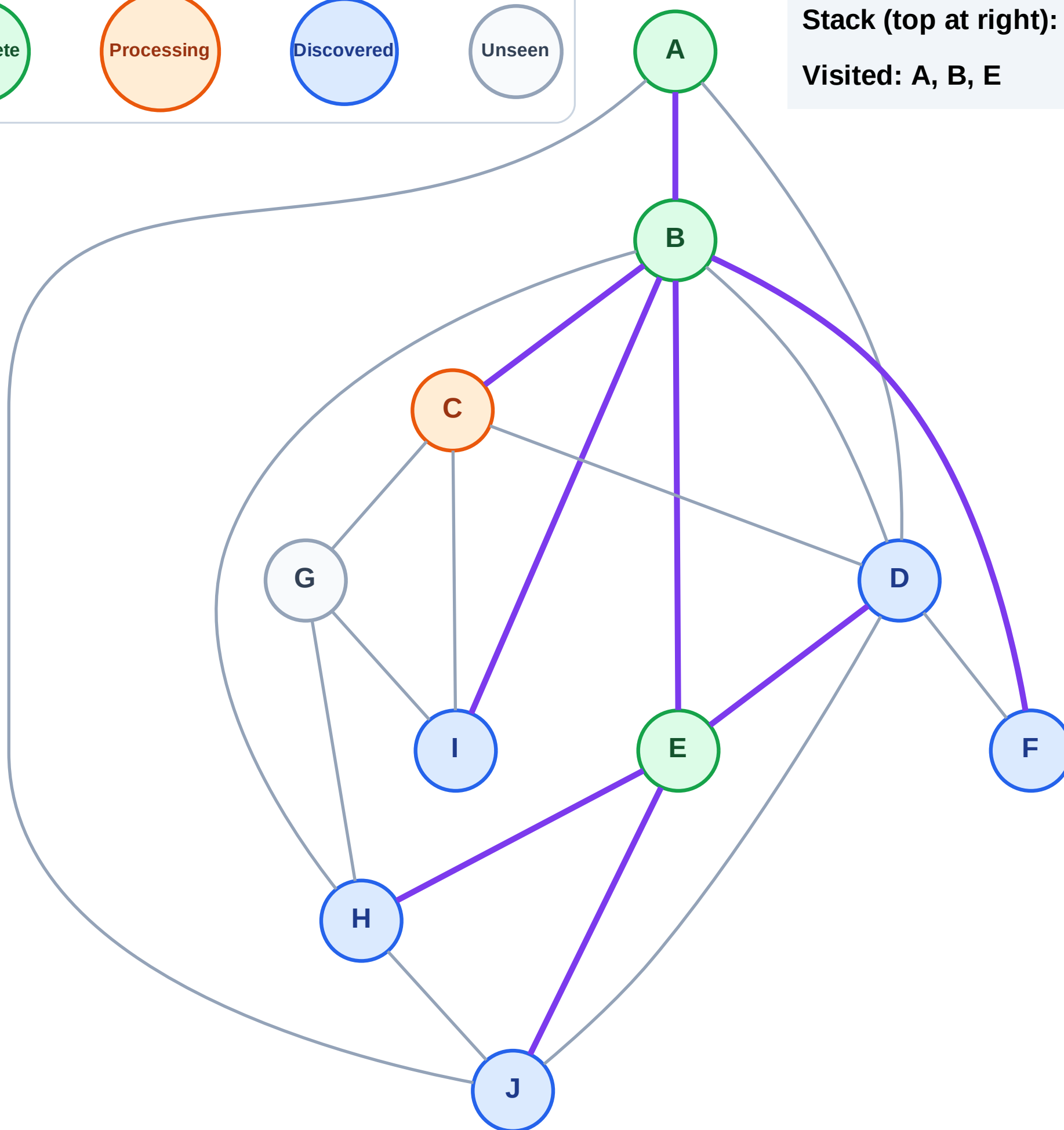
Discovered



Unseen

Stack (top at right): J | H | D | I | F

Visited: A, B, E



DFS Traversal

Step 9: Discover G from C

Legend

Complete

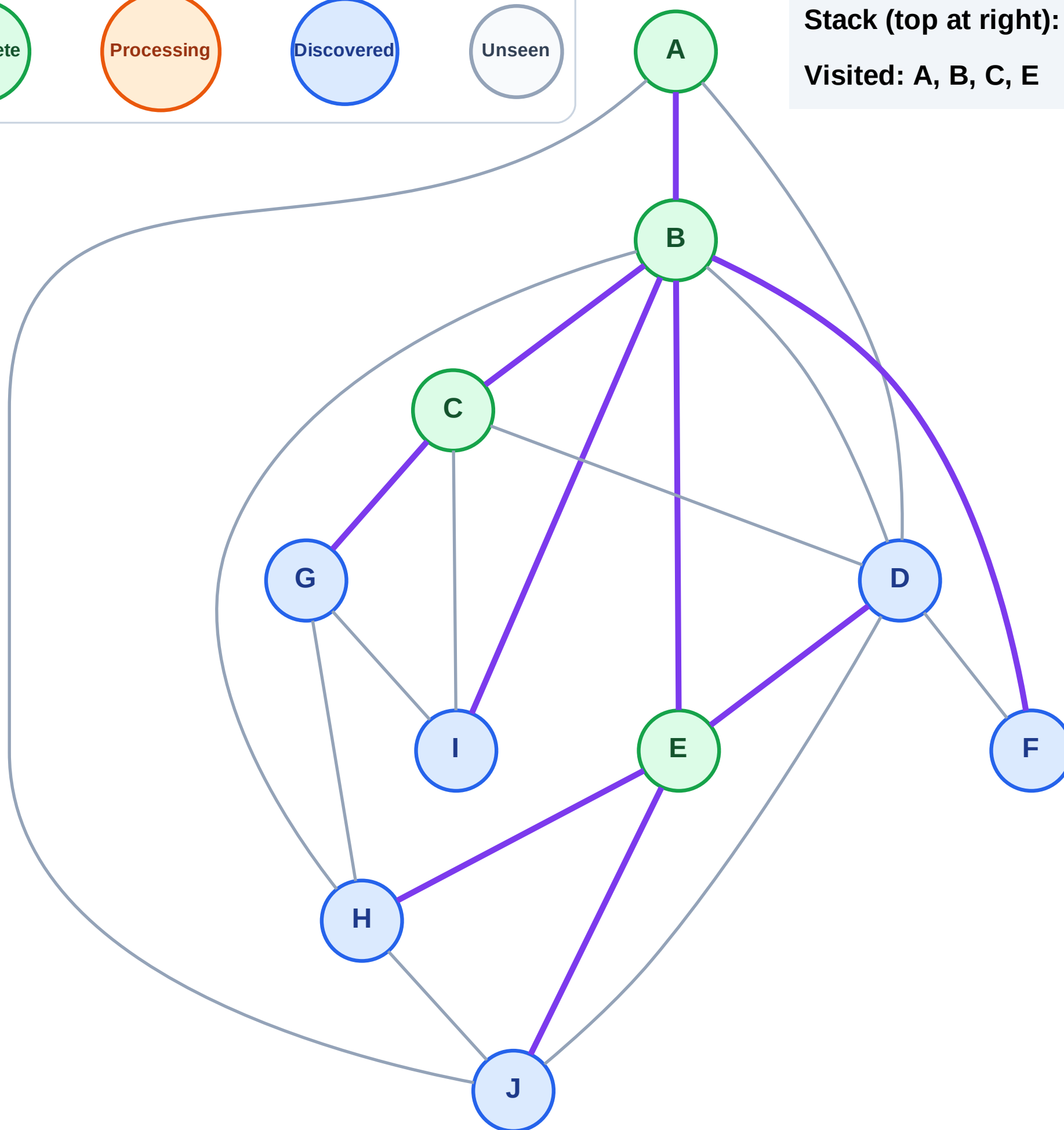
Processing

Discovered

Unseen

Stack (top at right): J | H | D | I | F | G

Visited: A, B, C, E



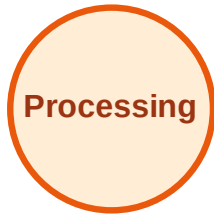
DFS Traversal

Step 10: Process node G

Legend



Complete



Processing



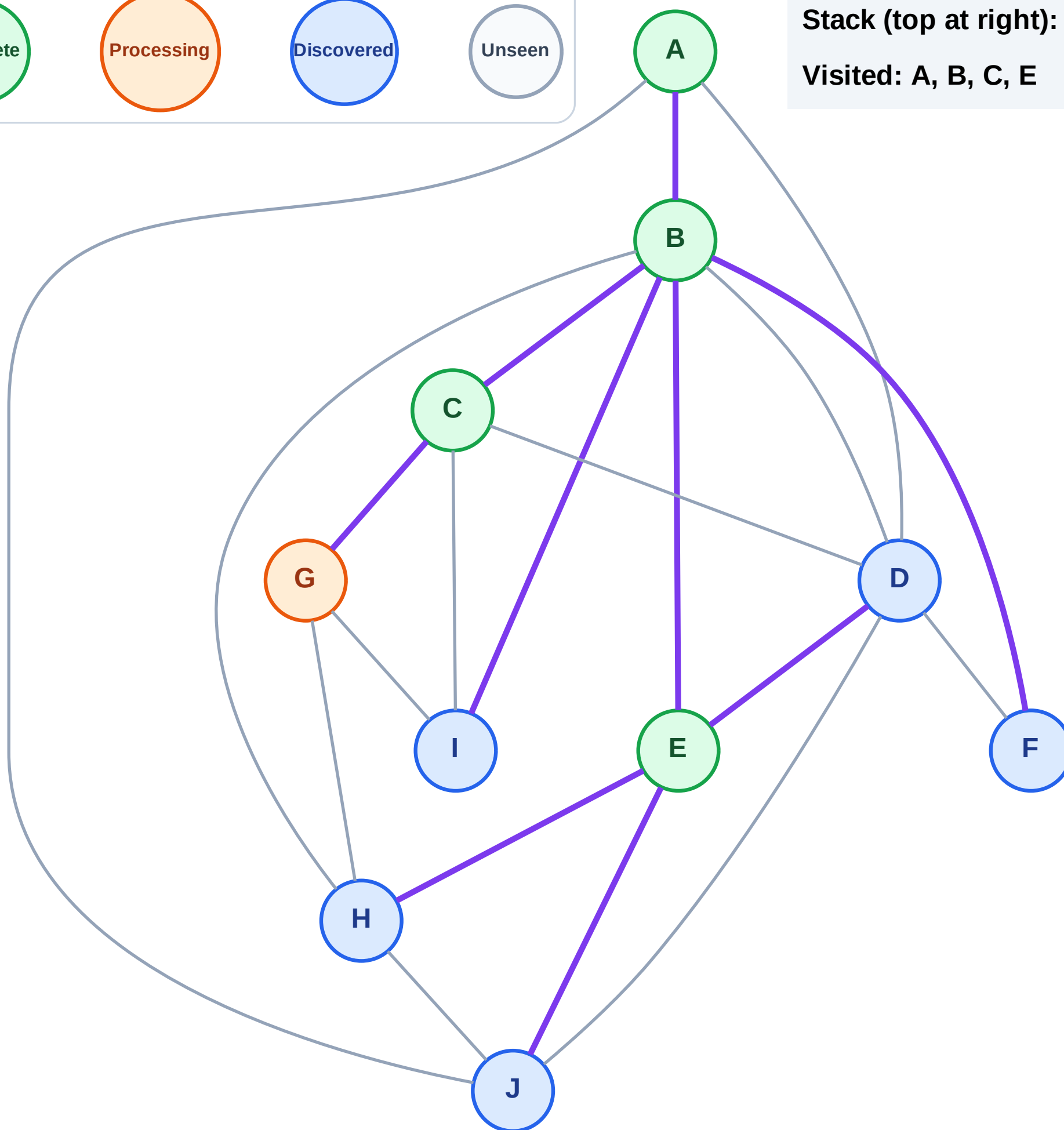
Discovered



Unseen

Stack (top at right): J | H | D | I | F

Visited: A, B, C, E



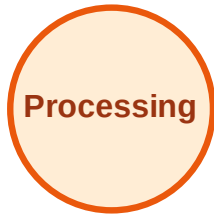
DFS Traversal

Step 11: No new neighbors from G

Legend



Complete



Processing



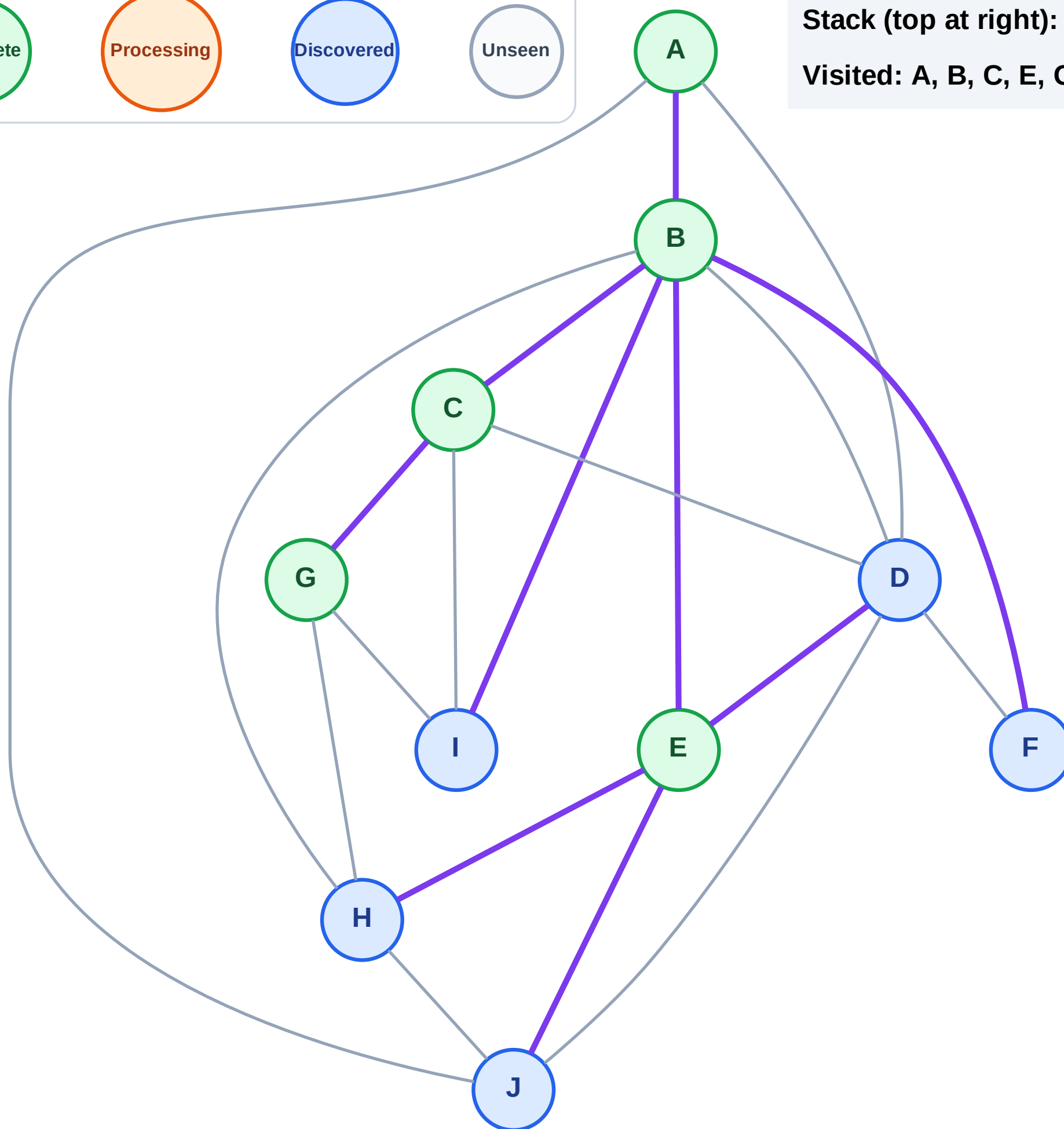
Discovered



Unseen

Stack (top at right): J | H | D | I | F

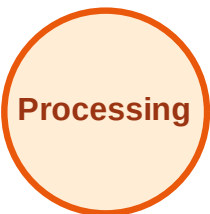
Visited: A, B, C, E, G



DFS Traversal

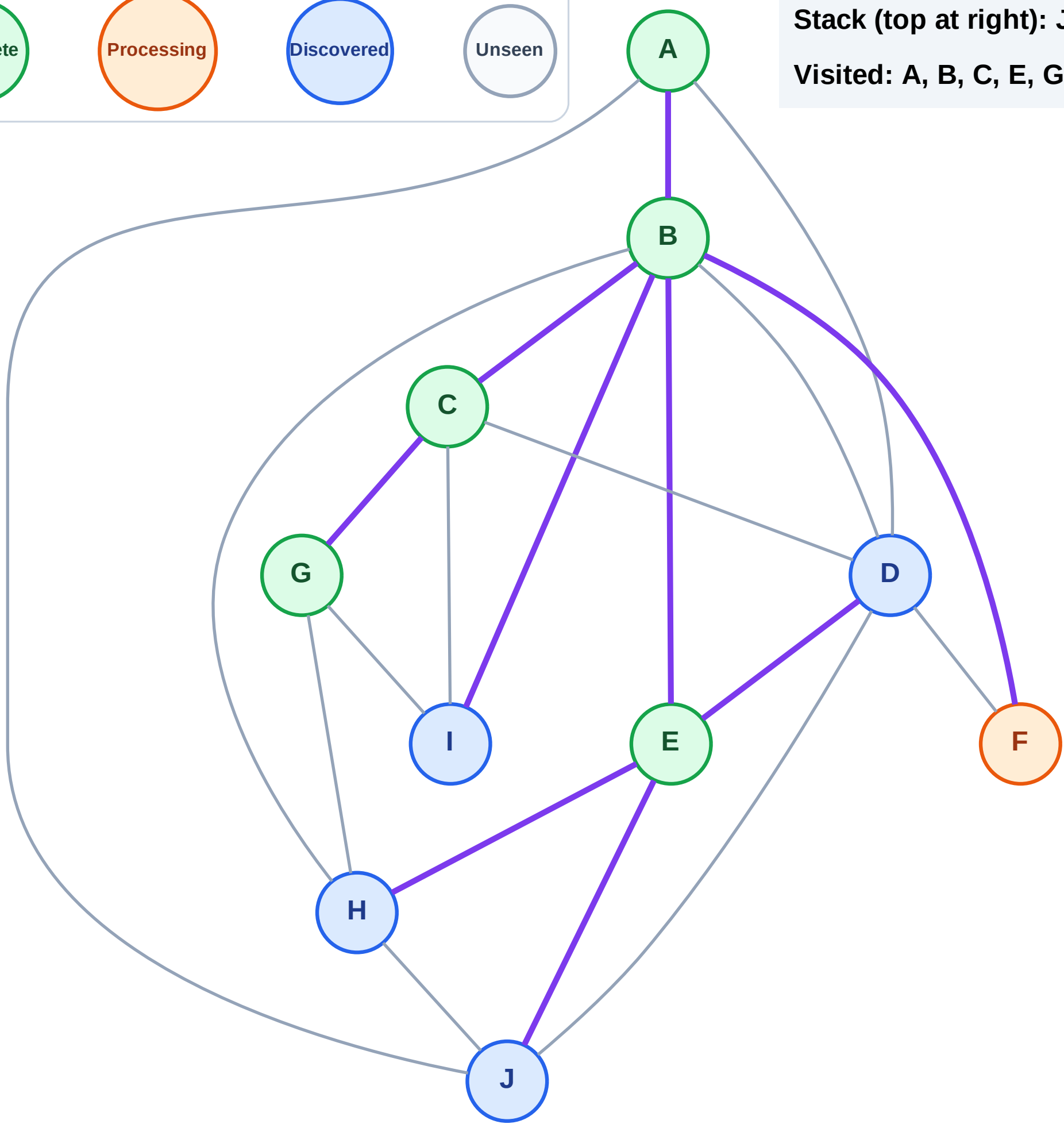
Step 12: Process node F

Legend



Stack (top at right): J | H | D | I

Visited: A, B, C, E, G



DFS Traversal

Step 13: No new neighbors from F

Legend



Complete



Processing



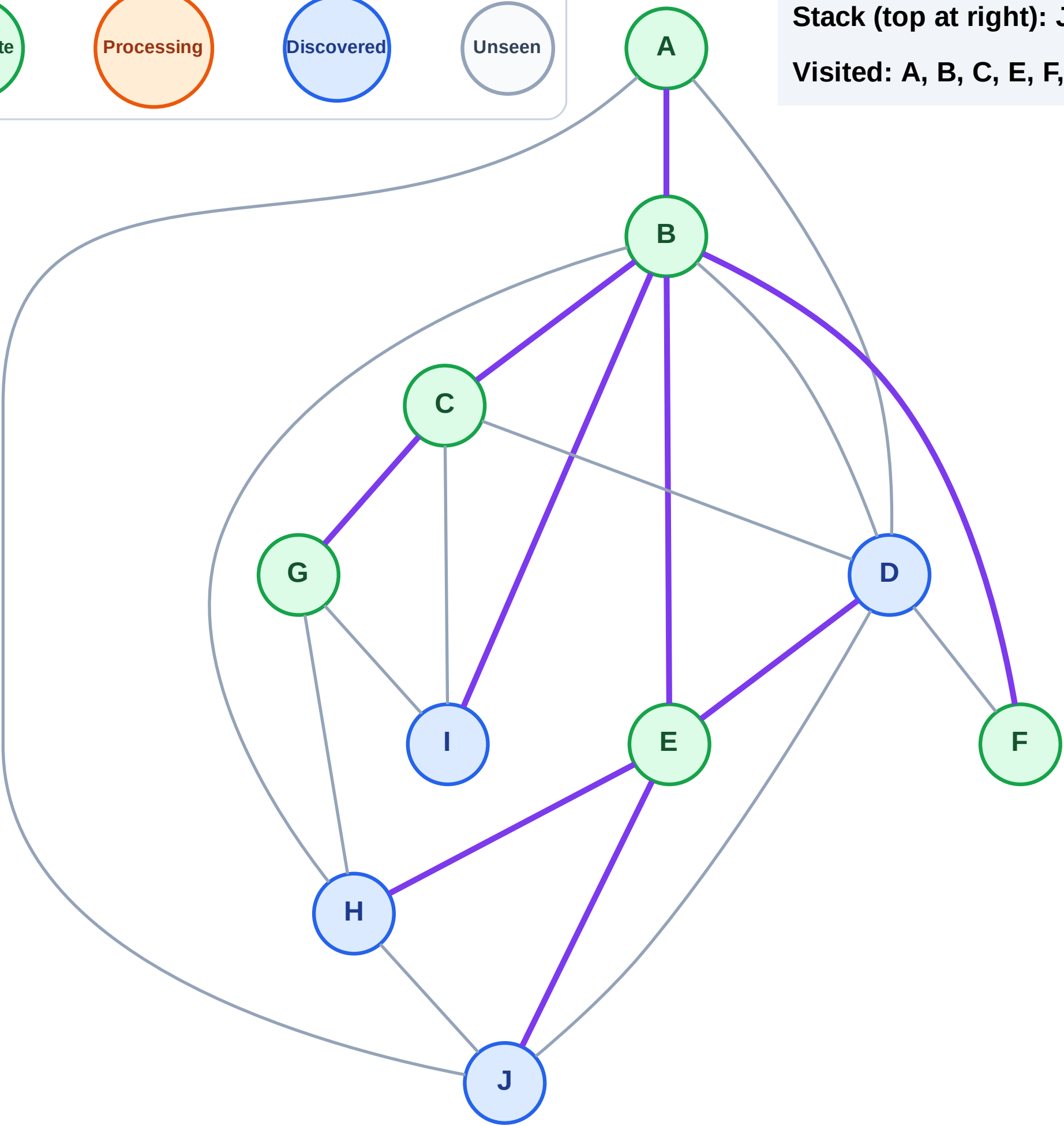
Discovered



Unseen

Stack (top at right): J | H | D | I

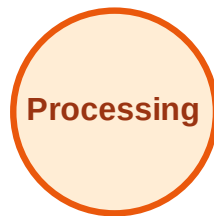
Visited: A, B, C, E, F, G



DFS Traversal

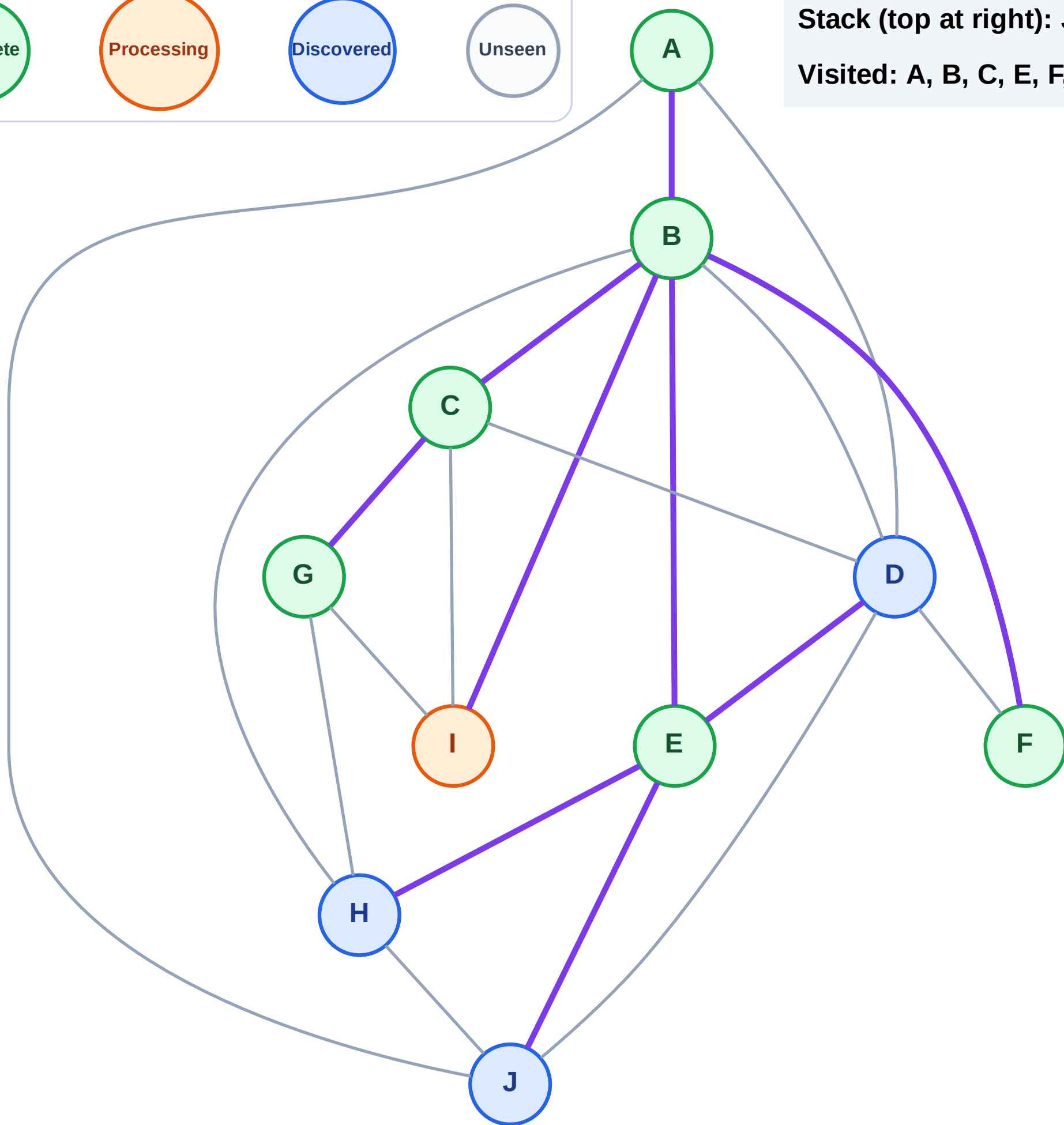
Step 14: Process node I

Legend



Stack (top at right): J | H | D

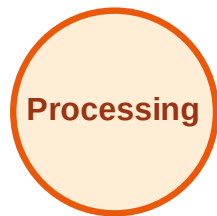
Visited: A, B, C, E, F, G



DFS Traversal

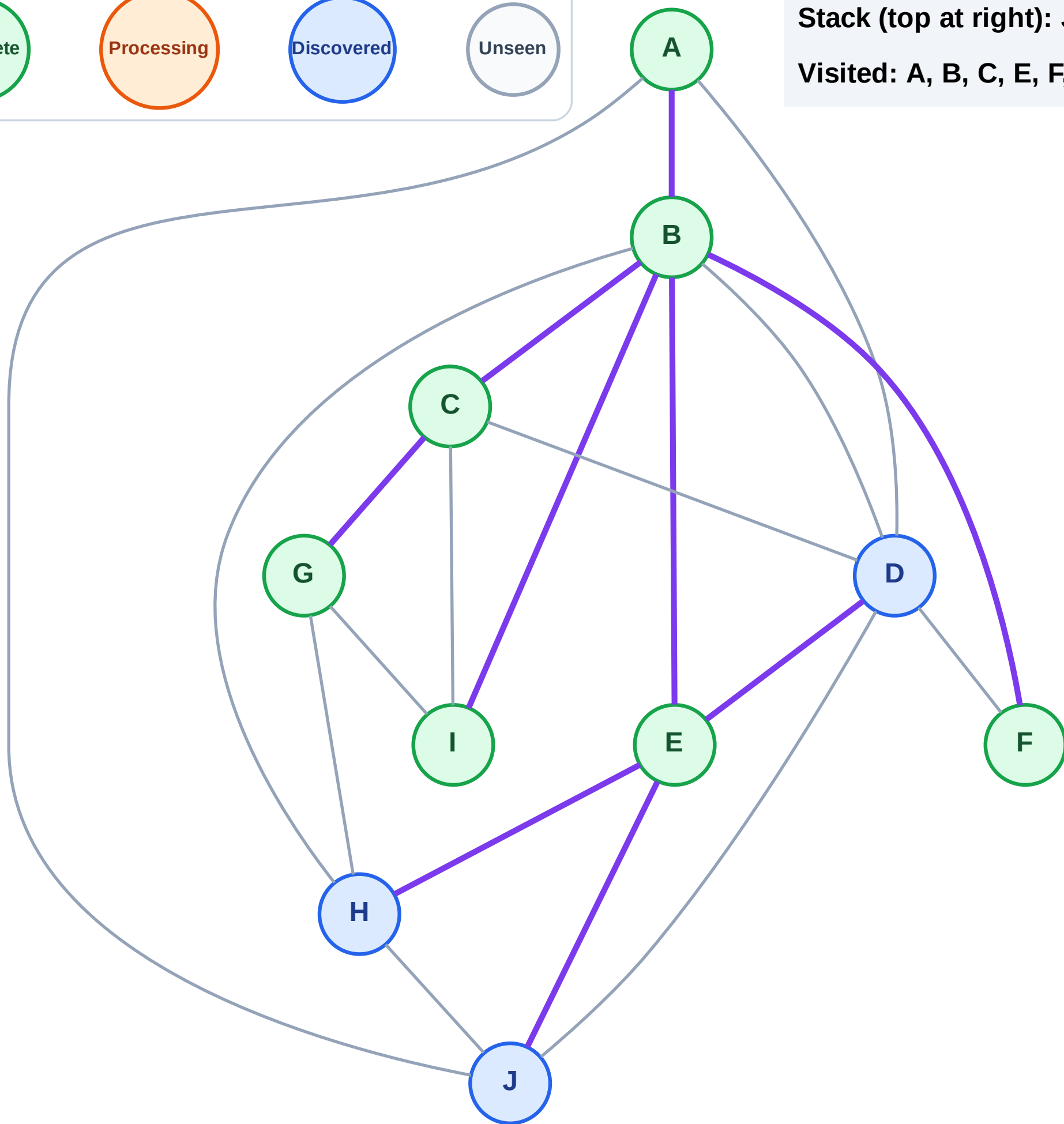
Step 15: No new neighbors from I

Legend



Stack (top at right): J | H | D

Visited: A, B, C, E, F, G, I



DFS Traversal

Step 16: Process node D

Legend

Complete

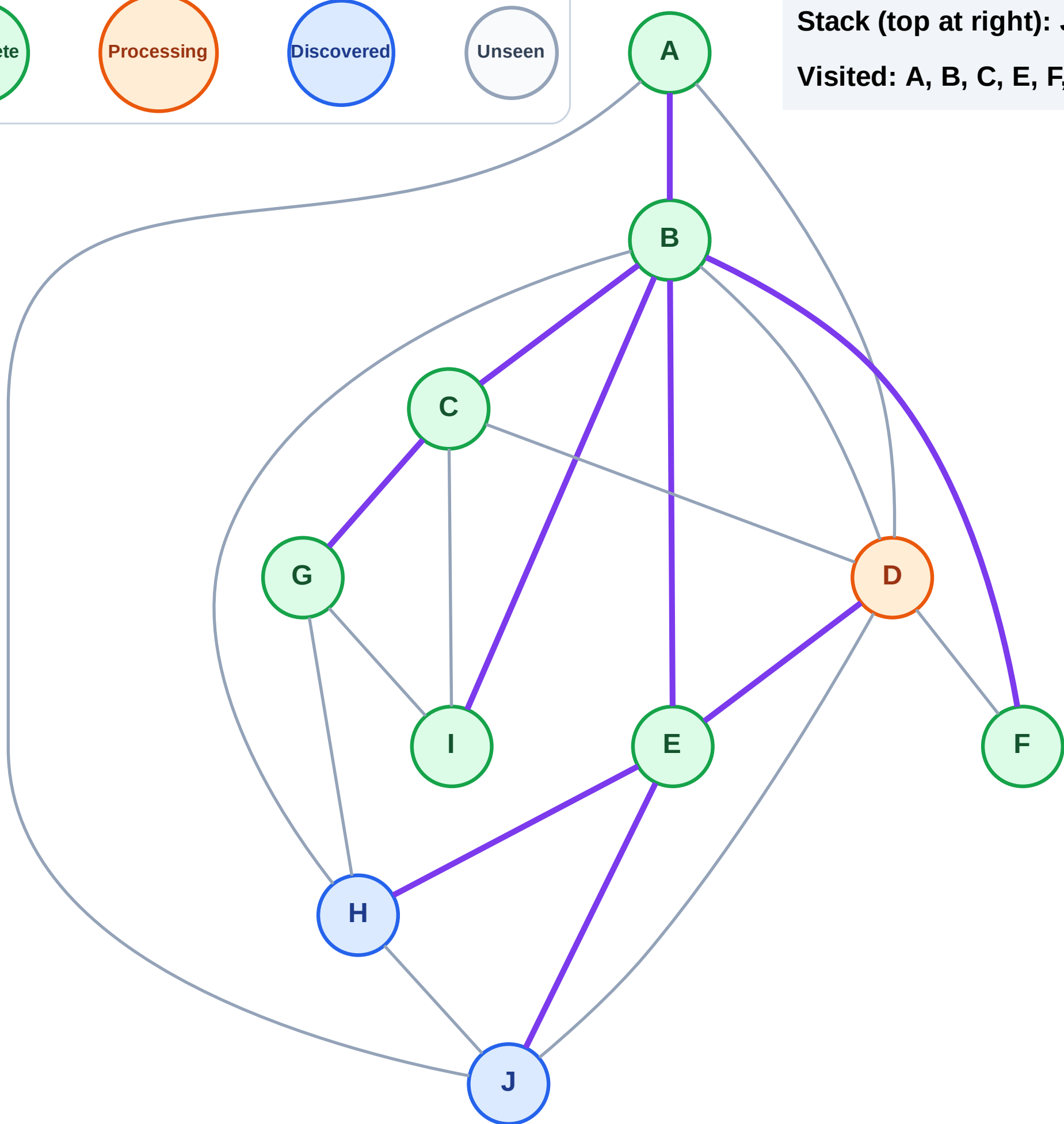
Processing

Discovered

Unseen

Stack (top at right): J | H

Visited: A, B, C, E, F, G, I



DFS Traversal

Step 17: No new neighbors from D

Legend

Complete

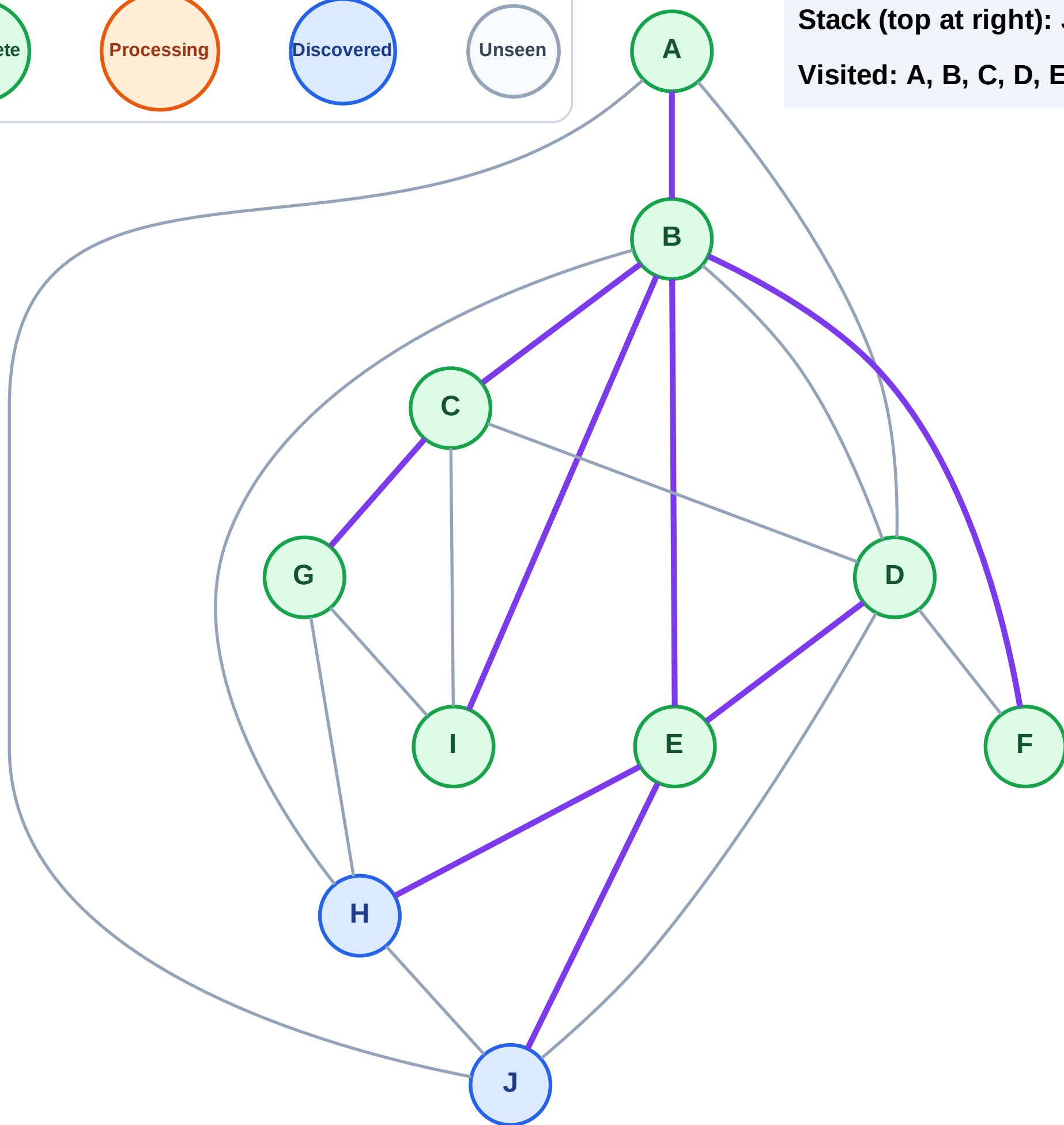
Processing

Discovered

Unseen

Stack (top at right): J | H

Visited: A, B, C, D, E, F, G, I



DFS Traversal

Step 18: Process node H

Legend

Complete

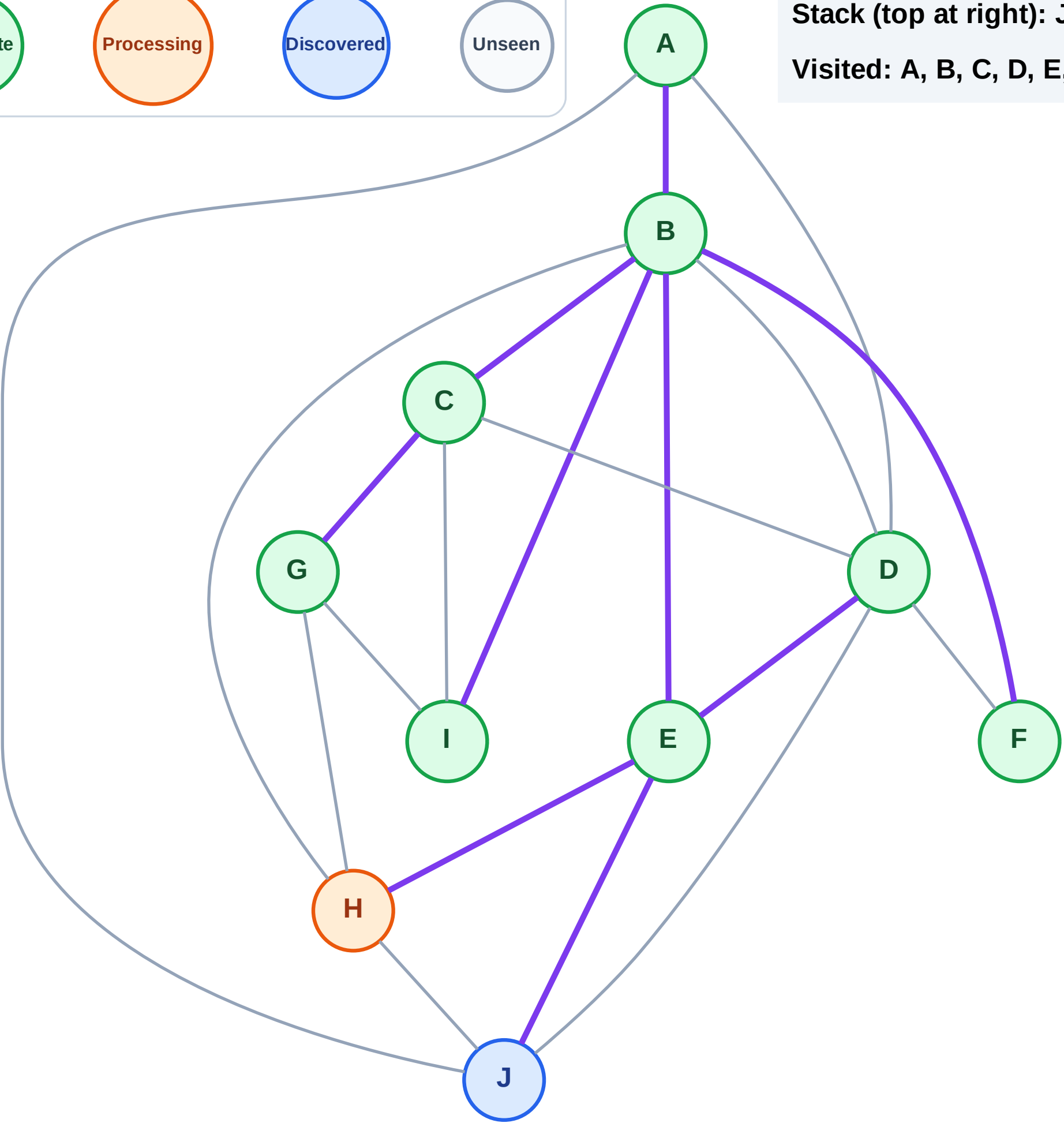
Processing

Discovered

Unseen

Stack (top at right): J

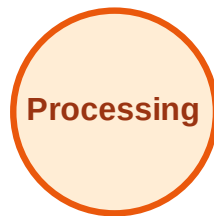
Visited: A, B, C, D, E, F, G, I



DFS Traversal

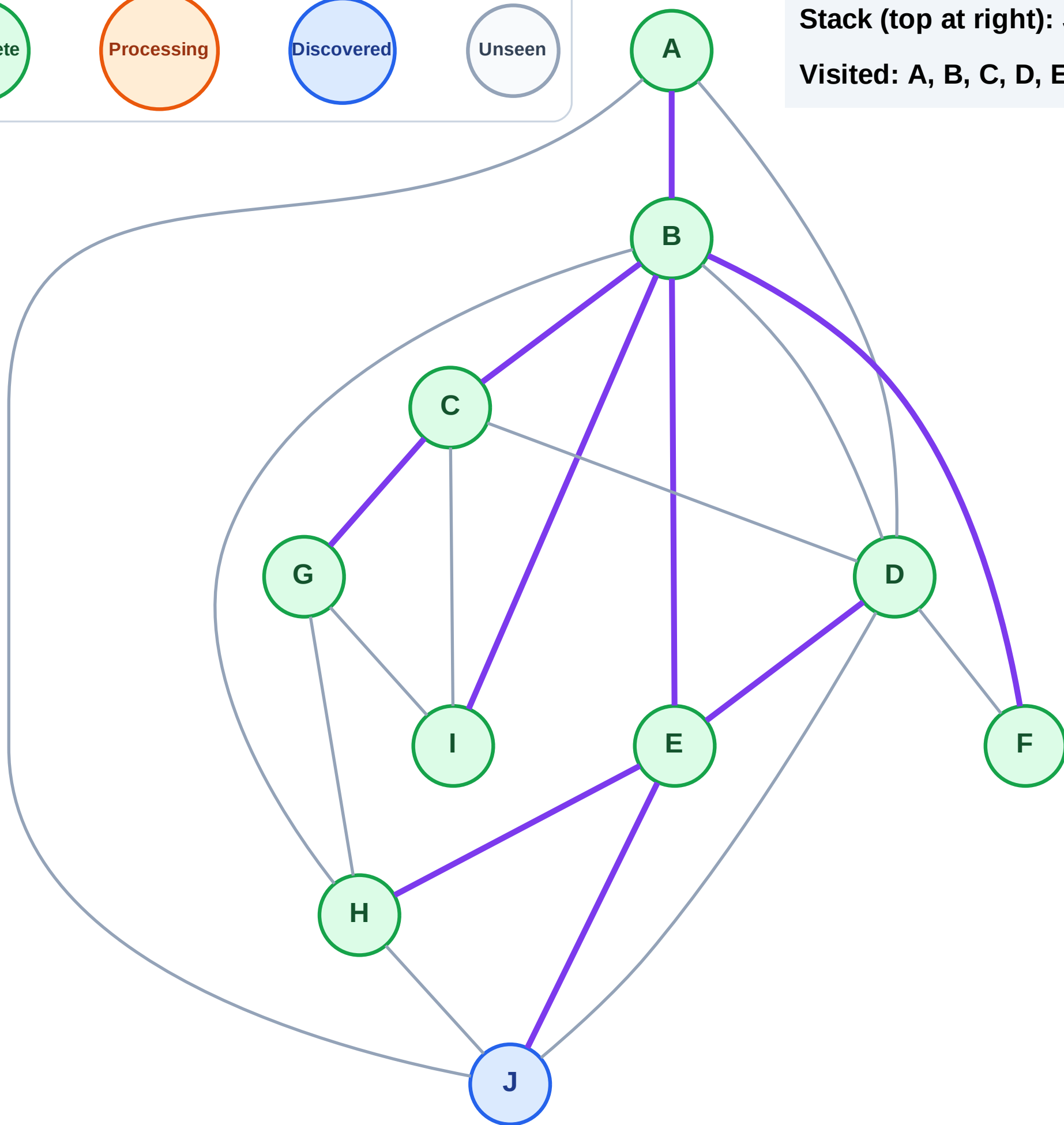
Step 19: No new neighbors from H

Legend



Stack (top at right): J

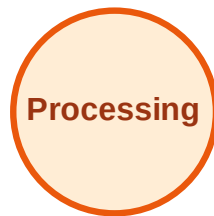
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

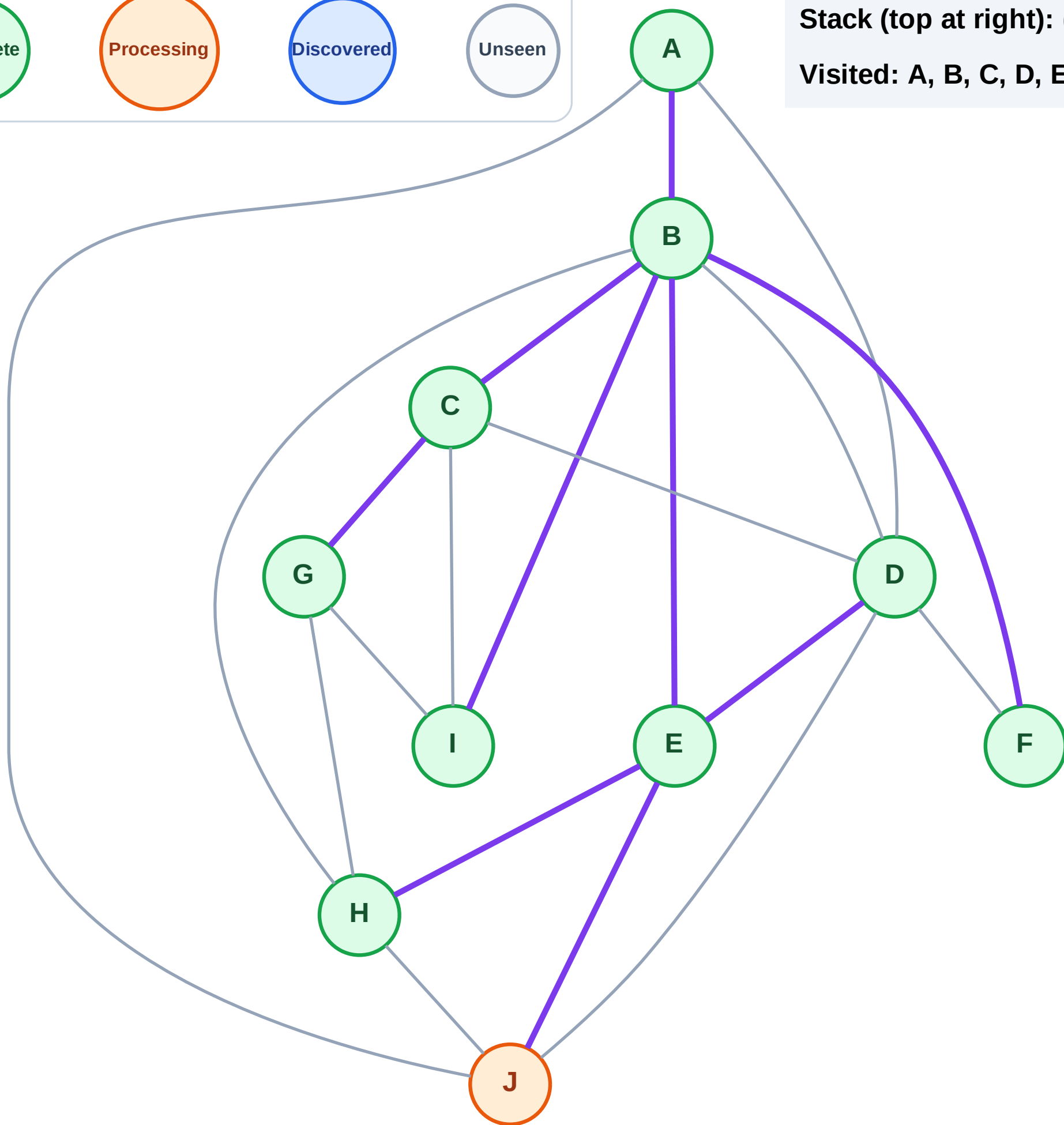
Step 20: Process node J

Legend



Stack (top at right): (empty)

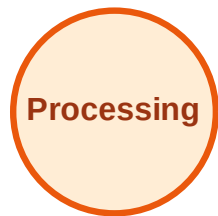
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

